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Management Science

Publication details, including instructions for authors and subscription information:
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To cite this article:
Zhuojun Gu, Ravi Bapna, Jason Chan, Alok Gupta (2022) Measuring the Impact of Crowdsourcing Features on Mobile App User Engagement and Retention: A Randomized Field Experiment. Management Science 68(2):1297-1329.
https://doi.org/10.1287/mnsc.2020.3943

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MANAGEMENT SCIENCE
Vol. 68, No. 2, February 2022, pp. 1297–1329
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Measuring the Impact of Crowdsourcing Features on Mobile App User Engagement and Retention: A Randomized Field Experiment

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Received: October 21, 2019
Revised: September 30, 2020; October 27, 2020
Accepted: November 16, 2020
Published Online in Articles in Advance: June 3, 2021

https://doi.org/10.1287/mnsc.2020.3943

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Abstract. The most commonly cited issues with mobile apps are low user engagement and retention levels. In this paper, we use a randomized control trial to test the efficacy of crowdsourcing on enhancing user engagement and retention in the context of mobile gaming apps. We examine two specific crowdsourcing features: the ability to submit content and the ability to access crowdsourced content. We find that the content submission significantly increases engagement and retention by reducing users' hazard of ending a session by approximately 11% relative to the baseline and reducing the hazard of abandoning the game app by 14%. In contrast, giving users the ability to access crowdsourced content has no significant effect on engagement but has a positive effect on retention by reducing the hazard of abandoning the game app by 13%. Surprisingly, we find that the interaction effect of these two crowdsourced features is negative on user engagement. Individually, the submission feature manifests itself via empowering users to control their product use experience, whereas the access feature's positive effect on retention is mediated by diversity and novelty of content. However, the two effects are not complementary. It turns out when these two features are given together, the empowerment enabled by one's own submission is crowded out by others' submissions, and this dominates the diversity benefit. Crowdsourcing features have heterogeneous impact on different user segments, with heavy users and users of longer tenure being more affected by the crowdsourcing features.

History: Accepted by Anandhi Bharadwaj, information systems.
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Funding: The research was supported by Carlson School of Management, University of Minnesota, Dean's Research Grant. However, it does not necessarily represent the views of Carlson School of Management or University of Minnesota.
Supplemental Material: The data files and online appendix are available at https://doi.org/10.1287/mnsc.2020.3943.

Keywords: crowdsourcing • online co-creation • user-generated content • mobile app • consumer empowerment • engagement • retention • autonomy • randomized field experiment

1. Introduction

Currently, mobile app use accounts for a staggering 15% of total Internet traffic, representing a huge market for app developers and companies. The number of mobile apps grew rapidly from 15,000 in 2009 to more than 5.09 million in 2019.[1] Despite a voluminous amount of mobile app downloads,[2] most apps are notoriously short lived in terms of their user lifetime. According to various mobile app marketing reports, the percentage of apps used only once is as high as 20%. Even for apps that are used more than once, more than 55% of these users do not use these apps frequently. Moreover, close to 95% of downloaded apps are abandoned by users within a month.[3] Specifically, mobile gaming apps, one of the largest app categories among U.S. users, face a relatively short user lifetime. A mere 10% retention rate a week after the initial download of a gaming app is considered a decent outcome. This suggests that app designers and companies must look beyond app download volume and consider user engagement and retention as core strategic goals. Given that app monetizing strategies can only contribute materially to the developers' bottom lines through the sustained use of their mobile apps, it is imperative for companies to induce user engagement and retention to boost the user lifetime value of mobile app products.

Concomitantly, opportunities involving the use of crowdsourcing to enhance user experience and

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loyalty have risen sharply in recent years. For instance, Starbucks (through MyStarbucksIdea) and Dell (through Dell IdeaStorm) have successfully deployed crowdsourced ideas to refine their product and service offerings. These attempts to involve users as co-creators of product/service design are motivated by a desire to sustain user traffic and increase loyalty to their brand. Moreover, organizations such as Wikipedia, Quora, Muji, LEGO, and Threadless (among others) have been successful at integrating crowdsourcing strategies into their core business operations.[4] The success of these organizations indicates that users place value on crowdsourced content and products.

In light of the issues faced in app use and the opportunities offered by crowdsourcing, we evaluate crowdsourcing-inspired design strategies in terms of their efficacy in enhancing user engagement and retention. We identify two subprocesses, namely product co-creation and content access, which characterize the distinct behavioral dynamics of crowdsourcing. The co-creation process allows individuals to submit their own content and ideas. According to the consumer[5] empowerment literature (Wathieu et al. 2002), product co-creation allows consumers to express their preferences on product features and thus shifts part of the control over the product to consumers, making co-creation a typical consumer-empowerment strategy (Fuchs and Schreier 2011). Co-creation gives consumers a sense of autonomy and ownership of the final product, which can be effective in enhancing engagement and retention levels (Füller et al. 2009). On the access side, products developed with crowdsourced inputs are likely to feature more diversity and novel ideas that could appeal to the end users (Jeppesen and Lakhani 2010). Thus, the process of accessing and consuming crowd-based content could potentially be linked to greater user engagement and retention (O'Brien and Toms 2008). Conventional wisdom suggests that the two processes complement each other in terms of enhancing user engagement and retention (Evans and Cothrel 2014).[6] In theory, their individual positive effects could create a virtuous cycle of interactivity and even a community amongst the game players as users would have shared experiences of molding and interacting with an IT artifact.

Despite the overall promising outlook of integrating crowdsourcing features, we are also aware that there might be user-related factors influencing the effectiveness of integrating crowdsourcing features in product/service offerings. Specifically, the high-quality crowdsourced content requires effort from the end users, posing an inherent risk for businesses in case that users from the community are dissatisfied with the provided crowd-based contributions (Jung et al. 2021). The success of the crowdsourcing model is critically contingent on having a crowd capable of providing content that appeals to the user community at large.

To investigate the causal impact of crowdsourcing features on user engagement and retention, we collaborated with a mobile gaming company to conduct a randomized field experiment on a "Catchphrase" game app. We created four versions of the app: (1) a control version without any crowdsourcing features, (2) a co-creation version that allows users to submit and play their own submitted content, (3) a content-access version that allows users to access crowdsourced content submitted by other users, and (4) a full-crowdsourced version with both word-submission and content-access features. We measure engagement and retention outcomes via game session length and time before churn. We then analyze how user engagement and retention differ between the treated versions of the app and the control app that has no crowdsourcing features. To understand the incremental impact of the subcrowdsourcing features, we further contrast outcomes of the full-crowdsourcing version with the versions with only one partial feature.

Our experimental results suggest that these crowdsourcing features can, in general, successfully address the challenge of low engagement and retention rates when integrated in gaming apps. We find that the content-submission feature significantly enhances user engagement, but the impact of content access on user engagement is not significant. Furthermore, user retention is enhanced by both the content-submission and the content-access features. Interestingly, we do not find complementary effects of these two crowdsourcing components on user engagement and retention. Rather, contrary to what one might expect, their interaction is negative.

Several nuances exist, calling for a mechanism-level understanding of the forces at work. A key novel insight from our causal analysis is that when given both options, users find their own content crowded out by content provided by others. This detrimental marginal effect of offering both features stems from a loss of perceived control and autonomy of the resultant product. The failure to see one's own content in a sea of crowdsourced content after hitherto being empowered to co-create results in unfulfilled expectations. Although content co-creation/submission is meant to give consumers more control or empowerment over the product and their use experience, it also, arguably, makes consumers more sensitive toward the level of control they have (Wathieu et al. 2002, Ascarza et al. 2016).[7] We find that the content-submission feature is responsible for increasing user engagement, but this is contingent on actual content submission as opposed to purely having the option to submit and experiencing an empowerment effect. In contrast, we find that user retention is enhanced by the mere exposure to the content-submission option. Finally, access to crowdsourced content is found to

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enhance user retention, which we find is driven largely by the mechanism of users having access to and engaging with specialized and niche content in crowdsourced environments. Together, these results suggest that creating and extracting consumer-side value from crowdsourcing is highly nuanced and requires a sophisticated understanding of the mechanisms at work.

To the best of our knowledge, we are among the first to demonstrate how crowdsourcing subcomponents can be used to enhance mobile user engagement and retention. Our research provides important theoretical and managerial implications for app companies, gaming platforms, developers, users, and academicians. Given the general efficacy of crowdsourcing in tackling low engagement and retention rates, app companies and developers should consider incorporating crowdsourcing elements within the design of their apps when possible. However, this should only be done with a detailed understanding of the underlying mechanisms. In this regard, the app designs deployed in our field experiment can provide guidelines for how individual crowdsourcing features should be used and integrated in mobile apps appropriately. Specifically, practitioners should be mindful that the user empowering co-creation strategy may collaterally alter users' mind status and their expectation of empowering benefit (Wathieu et al. 2002). Without effectively managing users' perceptions of empowerment by limiting their exposure to irrelevant crowdsourced content, adding the access to crowdsourced content to the co-creation option can be a great hindrance to users rather than enhancing their perceived use experience.

Our work also contributes to the literature on the antecedents of user engagement and retention in the mobile app context. Despite extensive literature on user engagement and retention, no studies to date have attempted to examine the impact of crowdsourcing elements on enhancing these key user metrics. Existing work has mainly focused on heightening these use measures from the perspective of customer relationship or service management (Bolton 1998, Berné et al. 2001, Gupta et al. 2006, Verhoef et al. 2010, Afeche et al. 2017) or through pricing, promotions, or advertising strategies (Lewis 2004, Iyengar et al. 2011, Lee et al. 2018). Findings from the focal study provide fresh perspectives on how app use measures can be improved by harnessing technological affordances of crowdsourcing, which were not attempted previously.

Additionally, we contribute to the literature on crowdsourcing. The extant literature on crowdsourcing has largely focused on examining its impact on the product-design stage and its deployment in contests/tournaments (Boudreau et al. 2011, Bayus 2013, Wooten and Ulrich 2017, Allen et al. 2018). It is largely silent on how a crowd-based strategy might influence user engagement and retention associated with consumer lifetime value. We shed light on the granular mechanisms via which crowdsourcing can affect user engagement and retention.

2. Literature Review

In our review, we first describe the relevant research on the mobile app market and the extant crowdsourcing literature. We then draw on literature from several related fields, including human computer interaction, marketing, and information systems, to provide a conceptualization of our key constructs and outcome measures.

2.1. Mobile Apps

Recognizing the need to monetize mobile apps, researchers have conducted a sizable body of research that examines the impact of pricing strategies on mobile app revenue. For example, Liu et al. (2012) empirically examine the impact of the freemium strategy on Google Play and find a positive association between the use of the freemium strategy and the app's sales and revenue. Through the estimation of consumer demand based on various app characteristics, Ghose and Han (2014) find that app demand varies with the monetization strategy used. In particular, monetization via in-app purchases increases demand, whereas monetization via in-app advertising decreases it. In a mobile reading app context, Zhang et al. (2019) find that a personalized pricing strategy based on the stage of user engagement can increase total revenue. Researchers in the healthcare area find that adding gamification features to glucose monitoring apps can improve users' use frequency (Cafazzo et al. 2012). This literature encourages us to explore how consumer-empowerment strategy through online co-creation would affect the user lifetime, which is an alternative strategy that may contribute to app profit.

Among human-computer interaction (HCI) research, Väätäjä et al. (2011) attempted to assess the type of mobile interface that is best suited to facilitate the process of content co-creation for news reporting. Zimmerman et al. (2011) evaluate different crowdsourcing prototypes to derive design requirements that could spur users toward content contribution so they could co-create bus arrival times with the transit service's bus schedule. Through a literature survey, Pan and Blevis (2011) seek to understand the role of interaction design in facilitating the development of crowdsourcing systems. Although early efforts in studying the use of crowdsourcing features in mobile apps, these studies aim to provide insights on the design strategies of integrating crowdsourcing features in apps and the usability consequences of such integration. To our knowledge, there is no existing

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empirical research that quantifies the impact of crowdsourcing features on downstream outcomes regarding the engagement and retention of mobile apps.

2.2. Crowdsourcing

Crowdsourcing is a digitally enabled strategy that explicitly involves consumers in the product-design process (Howe 2006), exemplified by open source software, open content production (e.g., Wikipedia), open innovation contests (e.g., Innocentive), and product co-creation (e.g., Starbucks).[8] In most settings, the crowdsourcing platform entails collaboration among users to solve problems or enhance existing products/services, through which users can use, evaluate, share, build artifacts, and network with other users (Doan et al. 2011). Based on the definition and existing deployment of crowdsourcing, crowdsourcing can be seen as two iterative processes that revolve around consumer-based inputs.

The first process is *product co-creation*, where users provide ideas and inputs for the product of interest and are directly involved in the product-design process. In the purest form of product co-creation, the user can specify inputs for her copy of the product and thus customize her product directly. Such co-creation that gives consumers more control and flexibility over the product or use experience is believed to inspire consumers with a feeling of autonomy, which promotes satisfaction (Benartzi and Thaler 2002, Wathieu et al. 2002). The crowdsourcing model extends this individual customization/co-creation model by allowing user inputs to be shared and implemented in a global copy of the product(s) that is then used by all consumers in the community.[9] For content-intensive products such as the game apps examined in this paper, user inputs mainly come from the submission of in-game content. Thus, in the gaming context, the product co-creation process allows users to co-create actual game content that constitutes an integral part of the product.

The second process, *content access*, pertains to the access of crowdsourced content as consumers use the focal product (or service). It is through this process that users benefit from products that are created via the integration of a wide variety of knowledge sources and user tastes. Content submitted by others can also generate a sense of community, especially when users resonate with the crowdsourced content submitted by other product users. In scenarios where the focal user can also submit content through product co-creation, she will be motivated by the awareness that her contribution may be potentially acknowledged by other users. For these reasons, the content-access process is an integral and necessary step for sustaining crowdsourcing models of operation, especially those that operate without financial incentives, which is typical of mobile apps. Thus far, our exposition alludes to why the processes of co-creation and content access are defining and integral aspects of crowdsourcing. Given the different driving mechanisms of these two processes to motivate users, it is highly pertinent to assess the respective effectiveness of these processes when applied to mobile apps.

The numerous applications of crowdsourcing in domains pertaining to knowledge creation have led to a strong focus on understanding the impact of crowdsourcing on idea generation and problem solving. Questions of interest in these studies include assessing the quality of crowdsourced solutions (Jeppesen and Lakhani 2010, Kazai et al. 2011, Poetz and Schreier 2012), evaluating the sustainability of the crowdsourced solution (Bayus 2013, Levine and Prietula 2013, Huang et al. 2014), and uncovering the optimal design of open tournaments (Kittur and Kraut 2008; Boudreau et al. 2011; Huang et al. 2012; Wooten and Ulrich 2015, 2017).

In a study looking at the quality of crowdsourced ideas, Poetz and Schreier (2012) show that crowdsourced ideas from users tend to outperform ideas generated from professionals in terms of novelty and customer benefit. In the context of open innovation, Jeppesen and Lakhani (2010) find that the effectiveness of crowdsourcing comes from the diversified knowledge of varied sources. Bayus (2013) studies the Dell IdeaStorm community and finds that individuals are unlikely to consistently submit high-quality ideas over time. Boudreau et al. (2011) explore how the number of competitors can influence user participation incentives and the quality of solutions, and Huang et al. (2012) investigate the optimal reward structure for these crowdsourcing contests. Researchers also examined the effect of feedback on idea generation (Wooten and Ulrich 2015, 2017) and how collaboration and coordination among crowdsourcing participants can affect group performance and content quality (Kittur and Kraut 2008, Ransbotham and Kane 2011, Ransbotham et al. 2012, Ren et al. 2015). Despite a rich set of studies on crowdsourcing, there are no known attempts to investigate the downstream consequences of deploying crowdsourcing features in the product-design phase. In particular, the relationships between crowdsourcing and user engagement and retention are poorly understood. Furthermore, the mechanisms through which crowdsourcing features affect user engagement and retention remain unclear, hampering our understanding of how crowdsourcing may be applied more broadly in the product-design process.

2.3. User Engagement and Retention

User engagement and retention are two core marketing indices that are positively associated (Webster

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and Ahuja 2006, Bowden 2009) key components of customer lifetime value. However, these two indices refer to different aspects of consumer behavior. While user engagement emphasizes the intensity of user participation, user retention focuses on the longevity of users' loyalty or interest toward a product. Theoretically, the measurements for user engagement and retention can diverge in a scenario where a user demonstrates a high initial engagement level but has a relatively short retention as a result of a rapid build-up of satiation under intensive use (McAlister 1982, McAlister and Pessemier 1982). Both use measures are commonly used to assess the customer lifetime value of apps to provide profit-related insights.

2.3.1. Engagement. The user engagement concept originated from two closely related fields: namely HCI (Chapman 1997, O'Brien and Toms 2008, O'Brien and Toms 2010) and marketing (Bowden 2009, Mollen and Wilson 2010, Verhoef et al. 2010, Brodie et al. 2011, Vivek et al. 2014, Calder et al. 2016). Within the HCI literature, user engagement is commonly defined as the interaction between the user and the system (Chapman 1997). O'Brien and Toms (2008) identify interactivity, perceived user control, variety, and novelty as important engagement attributes that characterize the interaction process. Notably, O'Brien and Toms (2008) characterize engagement as a process comprised of four distinct stages, namely point of engagement, period of sustained engagement, disengagement, and re-engagement, through which various distinct attributes of engagement are experienced. Users can and tend to cycle through the stages of engagement several times during a single use session, thereby making re-engagement intrinsic to the model. This conceptualization of engagement is holistic in that it considers the behaviors, cognitions, and emotions of users in the context of content, design, and interactive features of the application, instead of emphasizing solely what users are doing and thinking (Said 2004, O'Brien and Toms 2008) or feeling (Kappelman 1995). The marketing literature supports this view of engagement by acknowledging it as a multidimensional concept that goes beyond mere transactional involvement (Van Doorn et al. 2010) and is reflective of the deeper aspects of consumers' emotional and cognitive connections toward the product (Verhoef et al. 2010, Brodie et al. 2011, Vivek et al. 2014).

Although common engagement measures include psychological responses measured in consumer surveys (Kim et al. 2013, Calder et al. 2016), engagement behavior is highly specific to the context in which it is being observed (Brodie et al. 2011, Calder et al. 2016). This implies that appropriate measures of user engagement ought to be constructed based on the use scenarios involved. For example, in studies within educational psychology, user engagement is measured by the time spent on work, the concentration intensity, and the level of contribution of students (Klem and Connell 2004). In the context of online marketing, studies have measured customer engagement through the incidences of activities and actions indicative of heightened user involvement, such as the number of posted comments, likes, shares, subscriptions, site click-throughs, bookmarks, and emails (Ahuja and Medury 2010, Lee et al. 2018). Literature from HCI and marketing construes use frequency and intensity as integral parts of engagement, which are reflected by time spent using products or organizational offerings (O'Brien and Toms 2010, Vivek et al. 2014). Consistent with this conceptualization, Zhang et al. (2019) have measured user engagement in the mobile app setting via the app use time. Guided by the conceptualizations in extant literature, we measure user engagement through the duration of users' game sessions (more details given in a later section).

2.3.2. Retention. Retention is conceived of as a user lifetime value measure that contrasts with switching or churning (Chen and Hitt 2002). At its core, retention captures the longevity of users' loyalty to and interest in a product. Retention is largely measured by the duration of a customer's continuous relationship with a service provider (Bolton 1998, Thomas 2001). For example, in Thomas (2001), the retention process begins with the first purchase and continues until the termination of the relationship between the customer and the brand. In marketing literature, a major factor that determines customer retention is the extent of variety-seeking behavior (Berné et al. 2001), as the desire for change or novelty is related to the satiation that user has from consuming the product (McAlister 1982, McAlister and Pessemier 1982). Van Trijp et al. (1996) find that consumers with low product involvement demonstrate higher levels of variety seeking, that is, products with low perceived involvement, are less likely to retain their consumers. This is consistent with the aforementioned literature that views consumer engagement as an antecedent for customer retention (Bowden 2009).

Although most marketing researchers address use and retention separately, Ascarza and Hardie (2013) hold a holistic view of the two metrics and model the interdependency between them to achieve better predictive accuracy on consumer churns. Our research shares their holistic view by bringing the two metrics together when examining the impact of IT-enabled interventions on consumers' lifetime value.

3. Hypothesis Development

Although many studies have attributed the success of crowdsourcing models to the quality of crowdsourced content, few have looked beyond that to consider how

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the individual processes within crowdsourcing may contribute to the overall success of crowdsourcing. A critical understanding of the impact of each component within crowdsourcing is a necessary first step toward the appropriate design and integration of crowdsourcing features within products. Next, we discuss the impact of each of these crowdsourcing components and their joint impact.

3.1. Product Co-creation

By turning passive consumers into active co-creators, the process of product co-creation induces users to put effort into creating the focal product. Consequently, such investments can generate increased user commitment toward the product. Norton et al. (2012, pp. 453–460) termed this commitment as the "IKEA effect" and reasoned that the investment of effort in providing ideas to enhance the product can lead users to develop an attachment and love for the focal product. In a similar vein, Franke and Schreier (2010, pp. 1020–1031) refer to the "I design it myself effect" to explain why the co-creation process can result in a heightened commitment toward the product. From a task design perspective, co-creation provides users with greater autonomy and perceived control over the focal task and its outcome, which can in turn increase users' commitment to the task (Hackman and Oldham 1976). From the marketing perspective, co-creation is a consumer empowerment strategy, and shifting control to consumers usually results in improved choices and customer satisfaction (Benartzi and Thaler 2002, Wathieu et al. 2002, Wright et al. 2006, Fuchs and Schreier 2011).

Moreover, being involved in the creation of the final product, users tend to have an elevated valuation of the self-designed product that stems from a sense of accomplishment and psychological ownership of the product (Pierce et al. 2003). The increased levels of attachment and commitment induced by the co-creation process are likely to stimulate one's emotional and cognitive connections to the product, thereby driving user's engagement with the product. A similar relationship has also been proposed by studies in service research, which posit that the involvement of customers in the co-creation process can be an important antecedent for customer engagement (Verhoef et al. 2010). Moreover, by enabling users to provide add-on features or content enhancements through the co-creation process, users would be less inclined to seek out alternatives to the focal product because they know that they can customize the product based on their preferences. Accordingly, we propose the following hypotheses.

Hypothesis 1a. *Giving users a product co-creation option has a positive impact on user engagement.*

Hypothesis 1b. *Giving users a product co-creation option has a positive impact on user retention.*

3.2. Content Access

The second major process within crowdsourcing is content access, that is, the ability to view or use the crowdsourced content during product interactions. Crowdsourced content bears strong similarities to user-generated content (UGC), given that this content is typically provided by the community of users out of their own volition. Thus, insights from the UGC literature can provide insights into the possible impact of accessing crowdsourced content on user engagement and retention. Past works on UGC have found a positive link between review volume and/or ratings on product sales in various settings, including but not limited to books (Chevalier and Mayzlin 2006), movies (Duan et al. 2008), beauty products (Moe and Trusov 2011), and video games (Zhu and Zhang 2010). More recently, researchers found that information embedded in textual content provided by users can be a powerful driver of various market outcomes. For example, Ghose and Ipeirotis (2010) conclude that the subjectivity and readability of reviews can positively affect sales. Tirunillai and Tellis (2012) report an effect from the sentiment of UGC on stock trading outcomes. Goh et al. (2013) find that the persuasive effect of UGC is more than that of the marketers. Yang et al. (2018) show that engagement in terms of likes and comments depends on the valence and sentiment of the UGC post. Based on the existing set of works in UGC, it is likely that exposure to crowdsourced content can influence user attitudes, behaviors, and user engagement and retention. Access to the crowdsourced content is a crucial process in crowdsourcing, particularly for the gaming context, as the product of interest is of a content-intensive nature. Generated from a broad range of knowledge sources, crowdsourced inputs represent the synergistic integration of ideas across end users who introduce unprecedented diversity and novelty to the design of the product (Jeppesen and Lakhani 2010, Poetz and Schreier 2012). Hence, products designed with the crowdsourced inputs are likely of superior quality and novelty compared with those made by in-house developers (Nishikawa et al. 2013). As such, users are likely to perceive greater variety and novelty in the use of crowdsourced products.

By accessing the crowdsourced content, users experience greater product novelty. Such experiences can be engaging for the user, as novelty appeals to the user's sense of curiosity, encourages inquisitive behaviors, and promotes repeated engagement (O'Brien and Toms 2010). Additionally, products developed with crowdsourced ideas appeal directly to users' interests as these products evolve constantly based on the inputs provided by other users in the community.

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Moreover, because they have been contributed by actual product users, crowdsourced inputs would likely reflect relevant ideas that can improve product quality. According to information systems (IS) and marketing literature (Zheng et al. 2013, Ascarza et al. 2018), perceived content quality is the key to maintaining users' continued intention to use the platform. The constantly changing content in the crowdsourced product could potentially excite and interest users, thereby reducing their desire to seek variety through alternative substitutes. Accordingly, we propose the following.

Hypothesis 2a. *Giving users a content-access option (allowing users to view or use the crowdsourced content) has a positive impact on user engagement.*

Hypothesis 2b. *Giving users a content-access option (allowing users to view or use the crowdsourced content) has a positive impact on user retention.*

3.3. Full Crowdsourcing: Product Co-creation and Content Access

Full crowdsourcing embodies both of the aforementioned processes by simultaneously allowing users to take part in the product-creation process and to access crowdsourced content. Aside from the product co-creation and content-access mechanisms mentioned previously, the full-crowdsourcing process could enhance user engagement and retention through its role of building a sense of community for its users. Through crowdsourcing, users interact with other users through a series of use-related activities. After a piece of user-submitted content is released to the product community, peer users might generate further content enhancements in response to the contributor's work. The iterative act of product enhancement (via co-creation) and the use of crowdsourced content represent continuous episodes of implicit communication among users, wherein users interact with one another through their molding of a shared artifact (Evans and Cothrel 2014). This implicit communication mediated via the product artifact (i.e., the game content in our context) can enhance the level of interactivity experienced by users (Algesheimer et al. 2005), so constant product innovation through the incorporation of user-driven enhancements not only raises product quality but also has the effect of building a user community around the product (Gruen et al. 2000, McAlexander et al. 2002).

The community-building role of full crowdsourcing affects user engagement and retention around two different aspects. First, it introduces a social aspect to the product that enhances user engagement. According to Bonchek and Choudary (2013, pp. 1–2), "making a product social" is to connect people and products in a collaborative network. Having a social aspect embedded within products has the impact of increasing the value of the product and is shown to raise users' willingness to pay or seek deeper interactions with the product (Oestreicher-Singer and Zalmanson 2013). Such interactions brought by crowdsourcing would then further enhance user engagement. Second, users' attachment to the community that results from crowdsourcing enhances user retention. As users learn that their product-related inputs and contributions are accessed or adopted by other users they develop an attachment to the user community, which can act as a buffer against the tendency to churn and reinforces the willingness to continue product use. Evidence supporting the positive relationship between community building and user retention has been reported in the marketing literature (Gruen et al. 2000, McAlexander et al. 2002, Algesheimer et al. 2005). In sum, full crowdsourcing is likely to result in higher levels of user engagement and retention than when an individual suboption is given. As such, we posit the following.

Hypothesis 3a. *Giving users the full-crowdsourcing option has a positive impact on user engagement, and the magnitude of this impact is greater than when treating users with individual suboptions (co-creation or content access).*

Hypothesis 3b. *Giving users the full-crowdsourcing option has a positive impact on user retention, and the magnitude of this impact is greater than when treating users with individual sub-options (co-creation or content access).*

3.4. Heterogeneous Effects

At its core, both the co-creation and content-access options are features that allow users to experience a broader and more diverse set of product content not otherwise available through the content provided by companies and product developers. It is also likely that the benefits of these features are valued differently by different users given that there are individual differences in information processing abilities and preferences (Dillon and Watson 1996). According to HCI and psychology studies, these individual differences in abilities and preferences translate to unique user profiles to which different sets of needs are satisfied through the different use patterns. The consumer-empowerment studies in marketing also find that consumers' varying capabilities of solving co-creation tasks determine their perceived empowerment and enjoyment, which further determine their future participation (Füller et al. 2009). It is meaningful to understand the extent by which these crowdsourcing features enhance engagement and retention outcomes by identifying the user group that demonstrates the largest reactions to these features.

When considering the potential heterogeneous impact of crowdsourcing features, a relevant and natural way of segmenting users is based on their use intensity, as this characteristic relates to users' valuation of

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information and content. Past studies have found that heavy Internet users tend to gravitate toward specialized uses of the web and perform more information seeking activities (Assael 2005, Klein and Ford 2003). Such findings from a closely related group of users show the differences in psychological needs between the heavy and light app users. By engaging in the use of specialized online tools, heavy users are seeking out online affordances that provide them the autonomy to perform tasks. Based on this rationale, it is not difficult to see that these users are likely to have a preference for applications that could provide them with greater ownership where they are able to control outcomes of their use better. In addition, these users are likely to place greater value on the ability to access more content, given their tendencies to spend large amount of time searching for information online. Also, Wathieu et al. (2002) mention that there is a polarization effect among consumers' satisfaction toward the product when the consumer-empowerment strategy is used because different user segments may develop a less ambiguous evaluation of their experience as a result of increased involvement. Thus, it is quite plausible that the heavy users would appreciate the co-creation and content-access features more than casual users would, given that these features provide them with more autonomy and content. For this reason, we propose the following.

Hypothesis 4a. *The effects of the crowdsourcing features (i.e., co-creation and content access) have a stronger impact on the user engagement and retention of heavy users relative to casual users.*

Another meaningful user-segmentation characteristic pertains to user tenure. A common and key challenge faced by content companies is to keep existing users engaged, especially mature users who have consumed the product for an extended period. According to McAlister (1982), variety-seeking behavior is more likely to happen when the consumption of certain product attributes accumulates to reach the satiation point along users' consumption history. This implies that users who have consumed a sizeable amount of the content are more likely to have greater demands for content variety or novelty to sustain their interest. Thus, crowdsourcing features, which bring greater novelty to the product, will matter more to these users with a longer tenure. Both co-creation and content-access features can improve product novelty by either allowing users to create and add their own content or to access user-generated content created by the community. Thus, we posit the following.

Hypothesis 4b. *The effects of the crowdsourcing features (i.e., co-creation and content access) have a stronger impact on the user engagement of users with longer tenure relative to those with short tenure.*[10]

4. Experimental Context and Design

We chose to base our empirical investigation in the context of gaming apps for two reasons. First, gaming apps represents one of the largest app categories in the U.S. market, with an expected penetration rate of 64% in 2020.[11] Given the prevalence of gaming apps, our derived results are of practical relevance to a large set of app companies and developers. Second, despite their widespread adoption, gaming apps are notorious for having extremely low retention rates, with most apps losing 90% of users at the end of the first week after initial download. Our empirical effort is targeted at a segment of mobile apps in greatest need of alleviating these engagement and retention issues. Specifically, we chose to rely on a field experiment conducted via a social mobile game, which is an app version of the popular word-guessing game, Catchphrase. The gaming app is free to download and monetizes its business through in-app purchases of gaming content. In particular, users are spurred to purchase virtual card decks of in-game words to enhance their gaming experience. Each card deck contains hundreds of words and phrases[12] that are organized around a particular topic (e.g., "Springtime" or "Popular Brands"). The phrases contained in card decks constitute the main content of the gaming app, and the interestingness of these words is directly linked to users' engagement with the app. The objective of the game is to let the player's teammates guess the phrases that show up on the app screen within a time limit without saying a certain set of forbidden words. Players organize themselves into two teams, and the team with the greatest number of correct guesses wins the game. The nature of the game requires its players to be physically co-located to engage in a gaming session. Players in a game session play the game via one mobile device on which the app is installed and take turns to view game content on device as the game proceeds.

4.1. Experiment Design and Procedures

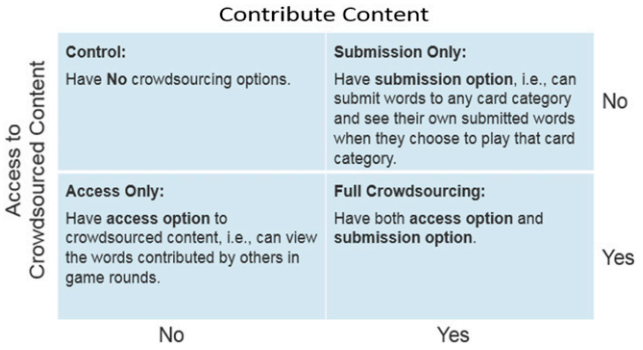
We partner with a gaming company to develop different versions of the Catchphrase gaming app. Each version of the app is designed to mirror the various crowdsourcing features discussed previously and deployed using a randomized controlled trial (RCT). In particular, we incorporate features that allow users to co-created product content by submitting phrases/words to the card decks and to access crowdsourced content by playing the game with words submitted by the community of gamers. More specifically, we operationalize the co-creation process as a content-submission function within the gaming app, and the content-access process as an in-game feature that allows users to access the crowdsourced content.

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Under this operationalization, we use a four-arm experimental design wherein the submission-only group, access-only group, and full-crowdsourcing group represent treatment groups to be contrasted with the control group, which is the version of the app without any crowdsourcing features (i.e., its word deck is provided by the app developers). The treatment groups are as follows: (1) The submission-only group can submit words to card decks and may view their own submitted words during the game play. We refer to the users' own submitted words as *Custom Words*. Users in this treatment group are unable to see the words submitted by other users, as this is a deliberate design to isolate the effects of crowdsourcing content access from the effects of product co-creation. (2) Users in the access-only group can view the words submitted by other players during their game play but cannot submit words or phrases to the game. Words submitted by other players are referred to as *Community Words*. (3) The full-crowdsourced version of the gaming app allows users to both submit words and view words submitted by themselves and other players during game play. (4) The control group uses the original version of the app with no crowdsourcing features. The four-arm design of our experiment is summarized in Figure 1.

To enhance the strength of our experimental manipulation, the app highlights any crowdsourced content during in-game sessions. For instance, in the submission-only and full-crowdsourcing versions, phrases or words submitted by the focal user bear the banner "Your Phrase" when they are being played in a gaming round. Similarly, in the access-only and full-crowdsourcing versions, words submitted by other users in the community bear a banner "Submitted Phrase" when they appear during the game. Crowdsourced words submitted by users are stored in the card deck together with the words provided by the app developer and are designed to show up randomly throughout the game. The game developers perform quality checks on all submitted words to remove typos and to ensure that these submissions are appropriately categorized in the right topics.

Figure 1. (Color online) Four-Arm Experimental Design



Furthermore, to ensure that users are able to perceive the word-submission feature in the full-crowdsourcing and submission-only versions, the app was designed to have a button labeled "Add Phrases" in a salient location on the screen just before the start of a game session. The purpose of this design is to allow users to perceive the option to co-create product content regardless of whether they submitted words to the game or not. For the full-crowdsourcing and access-only groups, the number of crowdsourced words submitted to a particular card deck is marked on the upper left corner of the card deck (see Appendix A.1). Again, we deliberately placed the crowdsourcing feature in a salient location so that users are aware that their card deck consists of crowdsourced content submitted by fellow gamers from the community.

Based on our research into the mobile game context, we find that the average gaming session and retention rate is quite low to begin with.[13] According to industry reports, app use rate is slowing as the industry has reached its saturation point. Many app categories are now growing at the expense of others, when before, all were growing in tandem. This indicates that apps have maxed out on users' time, which is finite. This is a problem for new app businesses since it would not be easy to compete with the popular incumbent apps for users' time, such as Facebook, Messenger, Google, Gmail, Instagram, Amazon, Apple Music, and others.[14] The literature in mobile gaming further indicates that mobile app retention is extremely low,[15] where a loss of 80% of new users shortly after their app download is an expected trend. For our study app, a non-blockbuster app, we expect that most users are single-time users; that is, drop-out rates will be significant as most users will only play the game once.

The key considerations in designing our sample size for this study first revolved around recognizing that we are likely to be in a rare-event scenario with respect to our outcome. We foresaw that word-submission behavior might not be prevalent among users, as the act of thinking of meaningful words to be used for the game requires substantial effort. To be able to perform analysis on word-submission behavior, we were considerate of the number of users to have sufficient instances of word-submission events. Furthermore, we attempted to alleviate concerns revolving around generalizability as much as possible. Last, to the extent feasible, we wanted to solicit a larger population of users so that the sample would be more representative of the entire population of app gamers. Thus, the derived results would also apply to other users who did not participate in the experiment. With these trends in mind and the need to

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consider the time and cost borne by our partnering company in the execution of a field experiment, we accumulated more than 750 users for each of the experimental groups over 113 days (from February 17, 2015 to June 9, 2015). By doing so, our study sample consists of approximately 1,125 game sessions, under the assumption that subjects play, on average, 1–2 game sessions before churning. This sample size of 1,125 sessions allow us to detect an effect size of 0.12 (i.e., small to medium effect size) at $\alpha = 5\%$ with a power of 90% for tests on user engagement. A sample of 750 users will allow us to detect an effect size of 0.16 at $\alpha = 5\%$ with a power of 90% for tests on user retention, which is considered reasonable.[16] We do not attempt to arrive at the exact number of game sessions and/or users to achieve an exact match with recommended effect sizes given in the literature (Cohen 1988), as the in-flow of users on any given day of the field study is unpredictable and these proposed values are in fact recommendations that do not require strict adherence.

Users are randomly assigned to each of the four groups. The random assignment happens at the instance when users install the game app, and users are assigned in a way such that the probability of being assigned to each of the four groups is equal. The random assignment of users to the treatment and control groups constitutes our main strategy against endogeneity issues that are otherwise present in the analysis of observational data. To eliminate systematic differences from learning and experience, we restrict our experimental sample to first-time users of the app. An additional benefit of relying on a pool of fresh users is that any difference in engagement or retention observed between the control and treated groups would not be attributed to the excitement gained from existing in-game features. Finally, we run a robustness test in Section 7 to examine and rule out whether contamination might have influenced the analysis results.

4.2. Outcome Variables

To measure users' engagement levels, we adopt the definition in O'Brien and Toms (2008), who view engagement as an integrative process of undergoing states of being engaged, disengaged, and reengaged. In our context, the *Session Duration* variable is an appropriate measure of user engagement as it captures the use level of the IT artifact and intensity of participation through the time spent on the focal task of interest.[17] Each active session in our gaming app begins from the time the user opens the app to the time she exits the app in a day, which can consist of one or multiple game rounds (see Appendix A.2 for more information on the game setting). In most cases, the act of exiting of the app can be understood as a loss of user interest in the game. As such, a game session in our context represents an instance of continuous game play where the users expend their energy playing the game till tedium outgrow users' interest to continue. Similar to the definition of O'Brien and Toms (2008), users can cycle through different engagement states as they play different games within the session. The more engaged users are, the more game rounds they play, resulting in longer session durations.[18]

To measure user retention, we look at the *User Lifetime* variable. Thomas (2001) defines customer lifetime as the time duration from the customer's first product purchase until her termination of the relationship with product or service provider. In our game app context, a subject's *User Lifetime* is the time elapsed between her initial installation and her last active date before abandoning the app. However, this simple measure is prone to right censoring because of the limited experiment period. To address the censoring issue, we assume that a user abandons the game app if a significant amount of time has lapsed since her last observed game session. Based on established literature on churning (Lewis 2004), the probability of a customer placing an order decreases as the time since the previous order increases. In a similar vein, we expect that the longer the time a user does not log in, the less likely she will return in future. In our data, we find that the interval between two game sessions for 96% of the users falls within 34 days. In other words, the probability of users being active after 34 days of nonactivity from their last observed session is very low. Based on the churn literature and the specifics of our context, a user is labeled as being censored if she is within the 34-day active window after her last observed game session at the end of our study period. Otherwise, a user is considered to abandon the game app after her last observed session.

5. Empirical Analysis

In this section, we first present the summary statistics of the data set and the model-free results based on the comparisons of sample means across different treatment groups. Next, with an eye on triangulation and to increase the precision of our treatment effect estimations, we deploy multiple relevant econometric specifications. Specifically, among other things, we treat our four-arm design as factorial and repeat our analysis and achieve exactly the same findings. Finally, we test the heterogeneous effect among different user sections.

5.1. Data and Descriptive Statistics

Our field experiment took place over a period of 113 days, during which 3,056 mobile game app users

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were randomly assigned to one of the experimental groups. The experimental assignment resulted in 757 subjects in the control group, 798 subjects in the full-crowdsourcing group, 742 users in the submission-only group, and 759 users in the access-only group. The field experiment yielded 36,938 game rounds spanning 4,440 game sessions.

We first check the efficacy of the randomization procedure. In particular, we are interested in understanding whether users assigned in each experimental group exhibit different pretreatment app downloading behavior or app use behavior. We do so by looking for differences in download rates by geographical location and the mobile devices owned by users (proxy by the app download store). We show the user demographics and present the results of our balance check in Table 1. The *t* test analyses suggest there are no statistical differences among the control and treated groups regarding users' geographic location and their app download sources. We also check for differences in the users' installation patterns over time. The monthly arrival pattern of new users appears to be similar between the control and treated groups, as evidenced in Table A.4.1 and Table A.4.2. Taken jointly, our analyses confirm that our random assignment of users has produced statistically comparable samples in each experiment groups.[19]

We further conduct a manipulation test (see Appendix A.5) to see if users in the treatment groups perceived the crowdsourcing features and understood how they work.[20] In particular, we check if users in the full-crowdsourcing and submission-only groups understood the word-submission feature and were using it in their game play. As reported in Appendix A.5, we find that 40 of 798 users in the full-crowdsourcing group and 36 of 742 users in the submission-only group submitted 171 and 130 words, respectively. Although- the proportion of users who submitted words is low, such a trend is expected and consistent with the participation inequality rule mentioned by Iyer and Katona (2015), wherein only 1% of users in the online community would actually contribute content. With approximately 5% of users submitting words through the full-crowdsourcing and submission-only groups (much larger than the 1% norm), this check suggests that users are indeed aware of the word-submission feature in their treatment and are using it in the game.

After assessing the validity of our experimental setup, we next report the summary statistics of users' engagement and retention levels in Table 2. We show our preliminary *t* test results on user engagement and retention of the field experiment in Table 3. As mentioned earlier, the two key measures of user engagement

Table 1. Summary Statistics and Mean Comparison of Full-Crowdsourcing, Submission-Only, Access-Only, and Control Group Users

	Mean (standard deviation)				*t* statistics (standard error)		
Summary statistics	Control (C) (N = 757)	Full-crowdsourcing (F) (N = 798)	Submission-only (S) (N = 742)	Access-only (A) (N = 759)	*t* statistics (F – C)	*t* statistics (S – C)	*t* statistics (A – C)
App source (proportion)							
Amazon App Store	50% (0.50)	49% (0.50)	50% (0.50)	52% (0.50)	–0.21 (0.03)	0.13 (0.03)	0.72 (0.03)
Apple App Store	10% (0.30)	10% (0.30)	9% (0.28)	7% (0.26)	–0.26(0.02)	–0.93 (0.02)	–1.92 (0.01)
Google Play	40% (0.49)	41% (0.49)	41% (0.49)	41% (0.49)	–0.38 (0.02)	0.43 (0.03)	0.38 (0.03)
Location (proportion)							
West-US	17% (0.38)	17% (0.37)	17% (0.38)	17% (0.38)	–0.19 (0.02)	0.04 (0.02)	0.05 (0.02)
Midwest/ Central-US	40% (0.49)	40% (0.49)	37% (0.48)	38% (0.48)	–0.01 (0.02)	–1.38 (0.03)	–0.93 (0.03)
East-US	33% (0.47)	34% (0.47)	35% (0.48)	35% (0.48)	0.55 (0.02)	0.99 (0.02)	0.99 (0.02)
Non-US	10% (0.29)	9% (0.27)	11% (0.31)	9% (0.29)	–0.68 (0.01)	0.65 (0.02)	–0.10 (0.02)

Notes. The first four columns of this table report the mean and standard deviation of the proportion of user installations from different downloading sources and geographic locations for the control and three treated groups. For example, the data set records users who download the game app from Amazon App Store as Amazon = 1, otherwise Amazon = 0. The mean value of Amazon equals 50% for the control group, which means that 50% of the user installation comes from the Amazon App Store, or the probability that an individual user in the control group installed the game app from the Amazon App Store equals 0.5. The last three columns are the results of the t-test calculating the mean difference between the control and the treated groups. F – C means difference between Full-crowdsourcing group and the Control group; S – C means difference between the Submission-only group and the Control group; A – C means difference between Access-only group and Control group.
*$p \le 0.1$; ** $p \le 0.05$; ***$p \le 0.01$.

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Table 2. Comparison of Full-Crowdsourcing, Submission-Only, Access-Only, and Control Group Users

	Session duration (in seconds)						User lifetime (in days)					
Experimental groups	Minimum	Maximum	Mean	Median	Standard deviation	Kurtosis	Minimum	Maximum	Mean	Median	Standard deviation	Kurtosis
Control (757 users)	5	1,1520	528.997	120	987.88	33.436	1	95	5.165	1	12.080	20.142
Full-crowdsourcing (798 users)	19	9,360	542.46	180	899.78	20.901	1	105	7.769	1	16.976	12.676
Submission-only (742 users)	30	8,280	656.291	180	1,139.004	14.402	1	109	7.391	1	16.318	15.343
Access-only (759 users)	9	10,531	542.800	120	967.229	25.031	1	108	6.771	1	15.850	17.096

and user retention are *Session Duration* and *User Lifetime*. The former records the time elapsed from the beginning of a session until the user exits the game session. The later details the time since a user's initial installation until her last observed session. The time spent on word submission is not included in the calculation of the *Session Duration*, such that this variable is a strict measure of game-play time.

According to the kurtosis values of the measures, we notice that the key use metrics are highly skewed. A game session of an average user usually lasts several minutes, whereas a few heavy players engage in their game session for hours. Similarly, an average user abandons their game app after their first use, whereas a few loyal players stick to the game for several months. A bootstrap simulation shows that the distribution of the sample means conforms to a normal distribution (Figure A.6.1), which indicates that t tests are appropriate for estimating treatment effects based on sample means.

By comparing the respective means of the treated groups with the control group (Table 2), we find that the average *Session Duration* is 2.5% higher for the full-crowdsourcing group, 24% higher for the submission-only group, and 2.61% higher for the access-only group. However, t tests in Table 3 show that only the session duration increment of the submission-only group is statistically significant ($t = 2.74$, $p < 0.01$). This result supports Hypothesis 1a, which states that the co-creation process has a positive impact on user engagement. But the engagement level of the access-only group is not significantly different from the control group, indicating that the Hypothesis 2a is not supported. We will explain why the full-crowdsourcing version, which includes the co-creation option, does not show a significant improvement in session duration in Sections 5 and 6.

In addition, the treated groups produced *User Lifetimes* that are longer than that of the control group. The differences in mean lifetimes are statistically significant. The mean user lifetimes of the full-crowdsourcing group, the submission-only group, and the access-only group exceed that of the control group respectively by 50% ($t = 3.47$, $p = 0.001$), 43% ($t = 3.01$, $p < 0.01$), and 31% ($t = 2.22$, $p < 0.05$).[21] The results support Hypothesis 1b and Hypothesis 2b, which posit that the submission and access features have a positive impact on user retention. These results are visualized as error bar charts in Figure 2 for ease of comparison. Details of the t test statistics are reported in Table 3.

To understand if the joint presence of the two crowdsourcing features produces better outcomes than the features offered individually (i.e., Hypothesis 3a and Hypothesis 3b), we contrast the full-crowdsourcing group with the submission-only and

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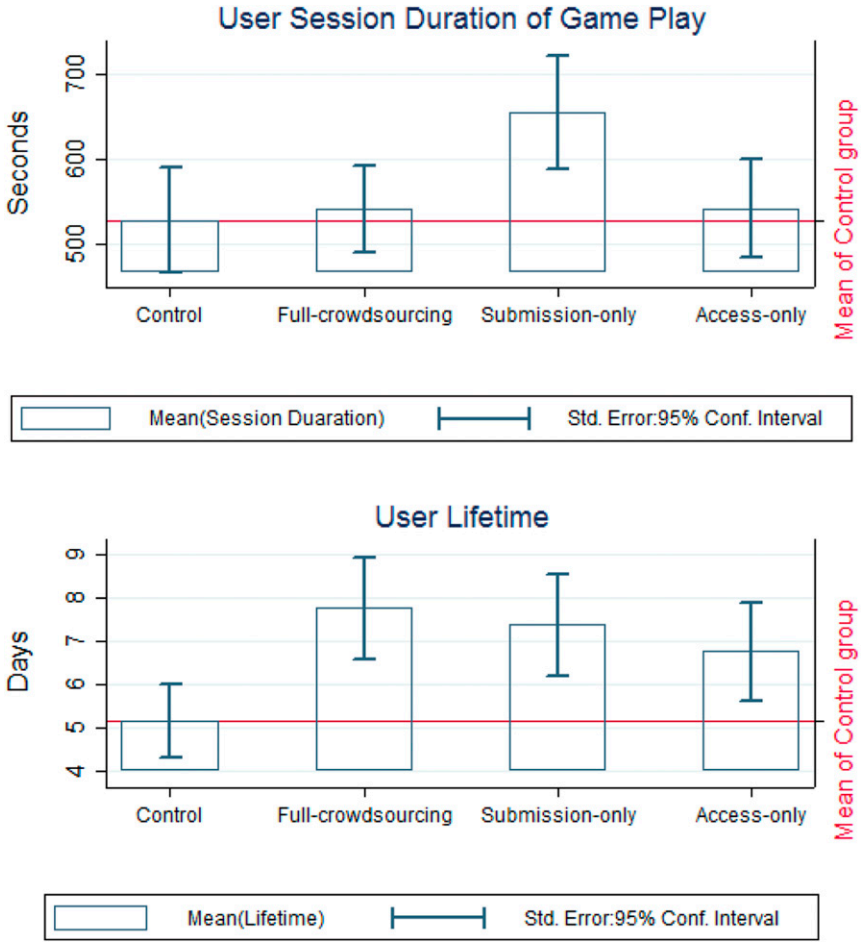
Table 3. *t* Test Results of Key Play Metrics Difference Between Treated and Control Group Users

Playmetrics		Full-crowdsourcing vs. Control	Submission-only vs. Control	Access-only vs. Control	Full-crowdsourcing vs. Submission-only	Full-crowdsourcing vs. Access-only
Session duration (in seconds)	Mean difference	13.460	126.719***	13.803	–113.258***	–0.343
	t statistics	0.335	2.736	0.324	–2.674	–0.009
	(Standard error)	(40.242)	(46.321)	(42.656)	(42.345)	(38.974)
	p value	0.738	0.006	0.746	0.008	0.993
	Observations (sessions)	2,204	2,145	2,103	2,337	2,295
lifetime (in days)	Mean difference	2.604***	2.226***	1.606**	0.379	0.998
	t statistics	3.469	3.005	2.218	0.446	1.198
	(Standard error)	(0.751)	(0.741)	(0.724)	(0.850)	(0.833)
	p value	0.001	0.003	0.027	0.656	0.231
	Observations (users)	1,555	1,499	1,516	1,540	1,557

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

access-only groups via *t* test analysis. Contrary to our expectation, the full-crowdsourcing condition tends to have worse engagement outcomes compared with the submission-only condition (Table 3). The engagement outcome of the full-crowdsourcing condition is not significantly different from that of the access-only condition or the control group. These two results indicate that Hypothesis 3a is not supported. A look at the analyses on retention outcomes show that the full-crowdsourcing group does not produce lifetime values that are significantly different from those in the submission-only and access-only groups, indicating that Hypothesis 3b is also not supported. We next execute formal tests using econometric models,

Figure 2. (Color online) Means of Key Outcome Variables Across Treatment Groups



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which permit us to dig deeper into understanding the mechanisms at work so that we can better explain the main effects.

5.2. Model Specification

5.2.1. Main Model. In addition to the model-free evidence presented previously, we next use ordinary least squares (OLS) as our baseline model and the Cox hazard model to further assess the validity of OLS results. The OLS model provides ease of interpretation of the coefficients, whereas the Cox model is highly appropriate for our setting as the model relates the time that passes before the occurrence of an event to covariates that may be associated with that quantity of time. Given that our outcome variables of interest are the amount of time spent in each game session (i.e., *Session Duration*) and the time in which users remain active (i.e., *User Lifetime*), the Cox model provides an intuitive way of describing how the hazard of game quitting (or churn) varies with different crowdsourcing features. Moreover, the Cox model is able to account for the data censoring issues common in the analysis of user retention (Bolton 1998).[22] Finally, the Cox model does not require the normality assumption (Cleves et al. 2008) and allows the baseline hazards to vary over time, which further makes it a fitting specification for our purposes. We first model the user outcomes as a function of a user's treatment condition and the relevant control variables. Specifically, the regression specification for contrasting various experimental groups is as follows:

$$session\,duration_{ij} = \alpha \cdot Treat_i + \beta \cdot Ctrl_{ij} + \varepsilon_{ij}, \quad (1)$$

$$lifetime_i = \alpha \cdot Treat_i + \beta \cdot Ctrl_i + \varepsilon_i. \quad (2)$$

For Equation (1), the use outcome is the session duration of user i at session j. Session duration is measured in seconds. In Equation (2), the outcome of interest is the observed user lifetime of user i and is measured in days. $Treat_i$ indicates the treatment group of user i, where this variable is operationalized as a set of three dummy variables each representing a crowdsourcing treatment condition. The reference group is the control condition. The engagement regression is conducted at the session level, and we include both users' demographic information and the session-level covariates to control for extraneous factors. The control variables of user i at session j is denoted as $Ctrl_{ij}$. On the other hand, the retention regression is conducted at the individual level and its control variables only include covariates at the user level. The control variables of user i is denoted as $Ctrl_i$. We do not include the session characteristics for predicting the hazard of abandoning the app because the effect of the session characteristics on users are temporary and would not be carried over to influence users' long-term decision on repeated playing. More specifically, we do not expect that covariates such as the gaming experience on the weekends or the number of co-players in the current round to influence the overall likelihood of playing the game in the future.

User i's demographic information includes her geographic location information and app downloading source. The geographic location dummies include *West-US*, *Midwest-Central* and *East-US*, with the *non-US* group serving as the baseline. The app downloading platform dummies include *Apple App Store* and *Google Play*, with the *Amazon App Store* group serving as the baseline. Geographic location can be important because users in different locations may be subjected to different population density and cultures of mobile game play behaviors. The app download source matters as it proxies for the phone and the operating system that the users use to play the game, which can influence the gaming experience.

Session characteristics affect user i's game play decision at session. They include *NumPlayer*, *Weekend*, and *Tenure*. *NumPlayer* indicates the number of players involved in a session. The relationship between session duration and the number of players is not necessarily monotonous. It is possible that the higher number of players, the more lively the atmosphere, which can subsequently enhance the user engagement level. However, since the continuation of the game session depends on the joint interest of all participants, it is also plausible that a higher number of participants may reduce the probability of reaching a consensus on initiating the next round. In this case, a large number of participants may decrease session duration. At the beginning of each session, users were asked to report the number of players under the three categories of *2−4 players*, *5−7 players*, *and 8+ players*, which serve as a prompt for users to self-organize themselves equally into two competing groups. In other words, the choice of number of players does not affect the setup of the game rounds, for example, the round duration or the number of words seen during the round. We use this as a covariate to control for any potential effects emanating from player size. *Weekend* is a dummy variable indicating whether the game round is played on a weekend. We include the weekend dummy as a control variable because game sessions that occur during weekend are likely to extend for a longer period of time given that users have a larger time budget. Finally, *Tenure* is the electronic *age* of a user at the current session. We measure tenure as the number of days elapsed since a user's first installation of the game. Players at different stages of their user life may demonstrate different engagement levels. For example, user engagement may first increase with tenure as users derive a sense of accomplishment after the initial process of familiarization but may

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decrease as tedium accrues to overweigh the sense of accomplishment. We include the *Tenure* covariate using a linear specification and include its squared term to assess the robustness of the results.

5.2.2. Survival Analyses. Although the OLS estimation results are easily interpretable, they are not immune to the data censoring problem. Because our observation of users' lifetimes may be right censored, especially toward the end of the study period, we triangulate our results and complement our OLS specifications with a set of analyses using survival analysis through the Cox hazard model. The game session duration and user lifetime can be considered random variables with a certain probability density function $f(t)$ and cumulative density function $F(t)$. The Cox hazard model estimates the hazard rate $h(t) = f(t)/(1 - F(t))$. The hazard rate $h(t)$ denotes the conditional likelihood that the game session or user life cycle terminates at time t, given that it has not happened during the interval $(0, t)$. Following the four-arm specification, the hazard of ending a session for user i at time t is specified as follows:

$$h(t \mid Treat_i, Ctrl_{it}) = h_0(t)\, exp(\alpha.Treat_i + \beta.Ctrl_{it}), \quad (3)$$

where $h_0(t)$ is the baseline hazard.

Like session duration, users' lifetime will be prolonged if users are treated with crowdsourcing features. We model the hazard of churn, or abandoning the game app at time t, as a function of users' treatment condition and other control variables. Ideally, time t is counted as the number of days elapsed since a user's installation. However, there might be a high proportion of the observations where multiple users abandon the game app on the same day, resulting in a loss of granularity on the outcome variable. To avoid this issue and gain efficiency in the estimation, we measure t in seconds so that there will be sufficient variation in users' failure time (i.e., time taken to abandon the game app). The hazard of abandoning the game app for user i at time t is modeled as follows:

$$h(t \mid Treat_i, Ctrl_i) = h_0(t)\, exp(\alpha.Treat_i + \beta.Ctrl_i). \quad (4)$$

5.2.3. Factorial Design Analyses. Although the previous model specification allows us to assess the main effects of content submission and access and the impact of the full-crowdsourcing condition relative to the control condition, it does not readily allow for the assessment of the marginal effect of content access on the content submission, meaning the interaction effect of these crowdsourcing components. In other words, we are unable to test if the joint presence of the crowdsourcing features produces better user engagement and retention than each of these crowdsourcing features could individually. For this purpose, we repeated our analysis using binary indicators to denote each type of treatment under a factorial design analysis. In particular, we estimate the following specifications:

$$\begin{aligned} session\,duration_{ij} &= \alpha_1 \cdot submission_option_{ij} + \alpha_2 \cdot access_option_{ij} \\ &+ \alpha_3 \cdot submission_option_{ij} \times access_option_{ij} \\ &+ \beta \cdot Ctrl_{ij} + \varepsilon_{ij}, \end{aligned} \quad (5)$$

$$\begin{aligned} lifetime_i = {} & \alpha_1 \cdot submission_option_i + \alpha_2 \cdot access_option_i \\ & + \alpha_3 \cdot submission_option_i \times access_option_i \\ & + \beta \cdot Ctrl_i + \varepsilon_i. \end{aligned} \quad (6)$$

The factorial equivalent specification of the hazard models is in the following format:

$$\begin{aligned} & h(t \mid Treat_i, Ctrl_{it}) \\ & = h_0(t)\, exp(\alpha \cdot submission_option_i + \beta \cdot access_option_i \\ & + \gamma \cdot submission_option_i \times access_option_i + \eta \cdot Ctrl_{it}), \end{aligned} \quad (7)$$

$$\begin{aligned} & h(t \mid Treat_i, Ctrl_i) \\ & = h_0(t)\, exp(\alpha \cdot submission_option_i + \beta \cdot access_option_i \\ & + \gamma \cdot submission_option_i \times access_option_i + \eta \cdot Ctrl_i). \end{aligned} \quad (8)$$

In this set of analyses, submission_option and access_option are marked 1 or 0 depending on whether the subject received submission and access features, respectively. In this specification, subjects in the full-crowdsourcing condition have submission_option = 1 and access_option = 1. We perform two sets of analyses here: 1) using the full sample of users and 2) using a subsample of users. An analysis using a subsample of users is conducted as a robustness check because the needed sample size in a factorial design analysis is smaller than that required in a four-arm design.[23] In our subsample analysis, we randomly sample the required number of users, perform the regression analysis, and repeat these two steps for 1,000 times. This procedure produces 1,000 sets of coefficients and standard errors, to which we take the average of these values as the results of the factorial design analysis.

On top of the factorial design analyses, we further examine the incremental impact of each additional crowdsourcing feature on the user engagement and retention via two further tests. Specifically, we contrast the full-crowdsourcing group with the submission-only group so that we can see the incremental effect of having access to crowdsourced words. We further compare the full-crowdsourcing group with the access-only group to tease out the incremental effect of having the word-submission feature.

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5.3. Estimation Results

5.3.1. Engagement. In this section, we first report the regression results of both the OLS model and the Cox hazard model under the four-arm experimental design. In models 1 and 2 of Table 4, we see that the submission-only treatment has a positive and significant impact on users' session duration. The treatment extends users' play time by approximately 126 seconds, on average. This result is consistent with the results from the Cox hazard models (models 3 and 4), which show that the submission-only treatment significantly reduces users' hazard of ending a session by approximately 11%. We find that neither the access-only treatment nor the full-crowdsourcing treatment significantly enhances user engagement. We further validate these results under a factorial design specification. Across all models in Table 5, we see that the submission option bears a positive and significant impact on session duration (lowering hazard rate of ending a game from 1 for the control group to approximately 0.89 for the submission-only group in the Cox model). These results mirror the model-free evidence in supporting Hypothesis 1a. To further understand the marginal impact of the access option on the submission option, we look at the interaction term of these crowdsourcing features in Table 5. Contrary to our expectation, the submission option and content-access option do not complement each other. In fact, the results show a significant negative interaction effect between the two crowdsourcing components, indicating that they are displacing each other. A closer look at the magnitudes of the interaction term and the

Table 4. Estimation Results of User Engagement Analysis Under a Four-Arm Specification

	OLS		Cox hazard	
	(1) Session duration (in seconds)	(2) Session duration (in seconds)	(3) Hazard ratio of ending a game session	(4) Hazard ratio of ending a game session
Treatment group				
full_crowdsourcing	13.460	12.808	0.967	0.966
	(42.835)	(42.508)	(0.041)	(0.041)
submission_only	126.719***	126.252**	0.887***	0.888**
	(43.338)	(43.012)	(0.038)	(0.039)
access_only	13.803	31.319	0.977	0.954
	(43.726)	(43.406)	(0.043)	(0.042)
Controls				
App source				
Apple App Store		234.094***		0.791***
		(52.113)		(0.042)
Google Play		59.148*		0.943*
		(31.614)		(0.030)
Geographic location				
West-US		73.359		0.901*
		(62.104)		(0.057)
Midwest-Central		152.902***		0.843***
		(55.678)		(0.048)
East-US		131.162**		0.850***
		(56.816)		(0.049)
Number of players				
5–7 *Players*		260.436***		0.846***
		(54.387)		(0.046)
8+ *Players*		132.465**		0.942
		(61.812)		(0.059)
Weekend		185.892***		0.835***
		(29.924)		(0.025)
Tenure		1.783*		0.997**
		(1.012)		(0.001)
Log likelihood			−33316.48	−33272.22
χ^2 (df)			9.05***	97.55***
			(3)	(12)
Observations	4,440	4,440	36,912	36,912
Subjects			4,439	4,439

Notes. For treatment, the baseline is the control group. For app source, the baseline is the Amazon App Store. For geographic location, the baseline is the non-U.S. users. For number of players involved in a round, the baseline is the option "2–4 Players." The tenure variable for OLS specification is log transformed as it is highly skewed. In the Cox hazard regression, "Subjects" records the number of sessions.

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

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Table 5. Estimation Results of User Engagement Analysis with Factorial Specification

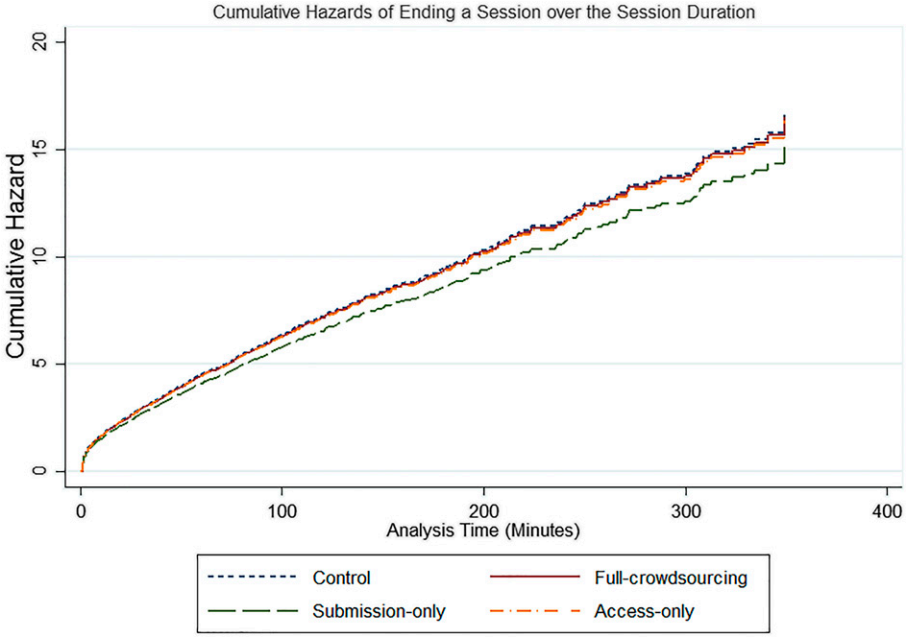
	OLS		Cox hazard	
	(1) Session duration in seconds	(2) Session duration in seconds	(3) Hazard ratio of ending a game session	(4) Hazard ratio of ending a game session
Treatment received				
submission_option	126.719***	118.838***	0.887***	0.888***
	(43.338)	(42.993)	(0.038)	(0.039)
access_option	13.803	26.230	0.977	0.954
	(43.726)	(43.373)	(0.042)	(0.042)
submission_option × *access_option*	−127.061**	−138.539**	1.115*	1.140**
	(60.252)	(59.675)	(0.067)	(0.069)
Controls added	No	Yes	No	Yes
Observations	4,440	4,440	36,912	36,912
Subjects			4,439	4,439

Note. The same set of controls from the main analyses are used in all specifications.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

submission term shows that the positive effect of the submission option is being negated by the interaction term, which explains why the full-crowdsourcing condition does not exert a significant impact on user engagement in Table 4. These results indicate that Hypothesis 2a and Hypothesis 3a are not supported, which is consistent with the model-free results. We discuss why this is the case in Section 6. To see the treatment effect on user engagement in a visually intuitive fashion, we use Figure 3 to depict users' cumulative hazard of ending a session across experimental groups (here, we report session duration in minutes for ease of interpretation). We see that only the submission-only group, represented by the long-dashed curve, demonstrates significantly lower hazard of ending a session than the control group.

5.3.2. Retention. We repeat the analyses above on user retention and report these results in Table 6 (four-arm) and Table 7 (factorial). In Table 6, models 1 and 2 present the impacts of the submission-only, access-only, and full-crowdsourcing treatments on the users' lifetime, and models 3 and 4 present the effects on the hazard of abandoning the game app. In the OLS models, we see that the three treatment groups (i.e., full-crowdsourcing treatment, submission-only treatment, and access-only treatment) have a significant and positive impact on users' lifetime (models 1 and 2).

Figure 3. (Color online) Treatment Effects on Cumulative Hazards of Ending a Session



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Table 6. Estimation Results of User Retention Analysis with Four-Arm Specification

	OLS		Cox Hazard	
	(1) Lifetime in days	(2) Lifetime in days	(3) Hazard ratio of abandoning the game app	(4) Hazard ratio of abandoning the game app
Treatment group				
full_crowdsourcing	2.604***	2.619***	0.830***	0.829**
	(0.783)	(0.781)	(0.052)	(0.052)
submission_only	2.226***	2.349***	0.860**	0.857**
	(0.797)	(0.795)	(0.055)	(0.054)
access_only	1.605**	1.733**	0.878**	0.868**
	(0.793)	(0.791)	(0.056)	(0.055)
Controls				
App source				
Apple App Store		2.702***		0.680***
		(1.009)		(0.058)
Google Play		−0.467		0.925*
		(0.588)		(0.044)
Geographic location				
West-US		2.648**		0.810**
		(1.132)		(0.073)
Midwest-Central		2.830***		0.859*
		(1.009)		(0.068)
East-US		0.673		0.778***
		(1.024)		(0.064)
Log likelihood			−14,369.89	−14,353.40
χ^2 (df)			9.86**	42.83***
			(3)	(8)
Observations	3,056	3,056	4,440	4,440
Subjects			3,056	3,056

Notes. For treatment, the baseline is the control group. For app source, the baseline is the Amazon App Store. For geographic location, the baseline is the non-US users.
*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

Correspondingly, these treatment conditions also have a significant effect toward reducing the hazard rate of abandoning the game app (models 3 and 4). Using model 2 as a reference, we see that the full-crowdsourcing treatment produces the largest increase in user lifetime, with an addition of 2.6 days (reduced the hazard rate of abandoning the game app by 17%), followed by the submission-only group, which increases lifetime by 2.3 days, and the access-only group, which increases lifetime by 1.7 days.

We further validate these sets of results under the factorial specification. Results from Table 7 show that

Table 7. Estimation Results of User Retention Analysis with Factorial Specification

	OLS		Cox hazard	
	(1) Lifetime in days	(2) Lifetime in days	(3) Hazard ratio of abandoning the game app	(4) Hazard ratio of abandoning the game app
Treatment received				
submission_option	2.226***	2.349***	0.860**	0.857**
	(0.798)	(0.795)	(0.055)	(0.054)
access_option	1.606**	1.733**	0.878**	0.868**
	(0.793)	(0.791)	(0.056)	(0.055)
submission_option × *access_option*	−1.227	−1.463	1.100	1.113
	(1.118)	(1.115)	(0.098)	(0.100)
Controls added	No	Yes	No	Yes
Log likelihood			−14,369.89	−14,353.403
χ^2 (df)			9.86**	42.83***
			(3)	(8)
Observations	3,056	3,056	4,440	4,440
Subjects			3,056	3,056

Note. The same set of controls from the main analyses are used in all specifications.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

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the main effects of submission and access options are both positive and significant in increasing user lifetime (reducing hazard of abandoning the app). The submission option increases the users' observed lifetime by 2.3 days (reduced the hazard of abandoning the game app by 14%), and the access option increases users' observed lifetime by 1.7 days (reduced the hazard of abandoning the game app by 13%). The results thus far indicate that Hypothesis 1b and Hypothesis 2b are supported, aligning with the conclusion gleaned from the model-free evidence. The interaction effects of these crowdsourcing components did not yield significant results in either the OLS model or the Cox hazard model. The results from Table 6 suggest that, although the full-crowdsourcing version resulted in significant increases in the raw count in user lifetime, this increase is not the result of a complementary effect from the submission and access component, but rather from the individual positive effect from each crowdsourcing component. These results indicate that Hypothesis 3b is not supported, which is congruent to what we have seen in the model-free results.

We illustrate the treatment effects on users' hazard of abandoning the game app through the cumulative hazards in Figure 4. In this figure, all three treated groups have significantly lower cumulative hazards than the control group along users' lifetime (user lifetime reported in days for ease of interpretation). Among the treated groups, the full-crowdsourcing group, represented by the solid curve, shows the lowest cumulative hazard of abandoning the game app, whereas both the submission-only group and the access-only group show significantly lower cumulative hazards of abandoning the app compared with the control group.

As a robustness check for the factorial design analysis, we repeated these analyses using subsamples derived from 1,000 sets of randomly drawn users, to which separate regressions were performed on each set of drawn users. Averaged values of the coefficients and standard errors of this analysis are reported in Table B.1.1. We see that under the reduced sample size, the results are qualitatively similar to those derived under the same analysis using the full sample in terms of sign and statistical significance.[24]

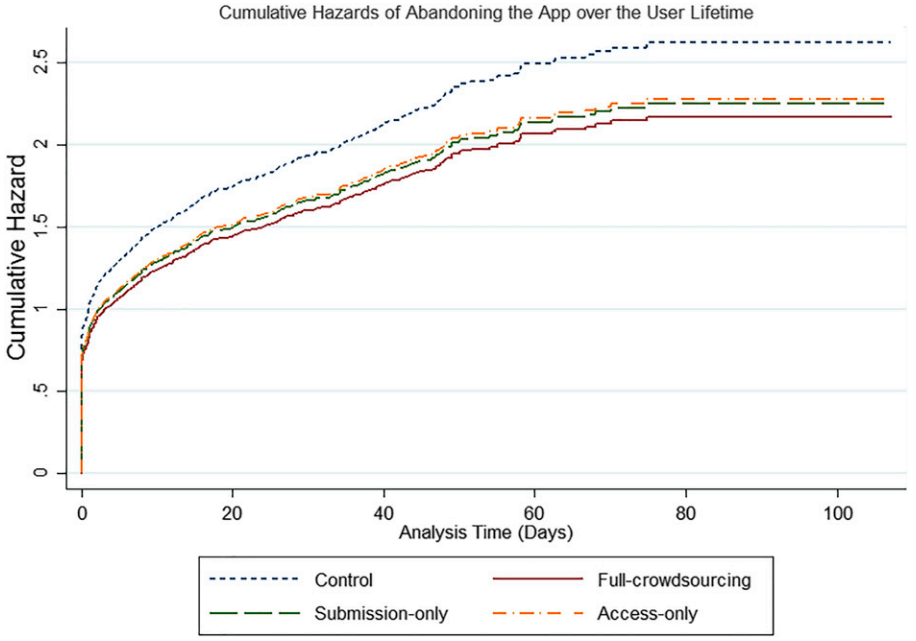
Finally, we supplement the factorial analysis results with a set of tests that contrasts the users in the full-crowdsourcing group with the users in treatment conditions that only have one crowdsourcing feature, that is, submission-only and access-only, to assess the incremental effect of access to community words and the option of word submission, respectively. In model 1 of Table B.1.2, we see that having access to the community-content feature on top of the word-submission feature has a negative effect on user engagement (increasing the hazard of ending a game session to be greater than 1). Specifically, the full-crowdsourcing group performs significantly worse compared with the submission-only group in terms of user engagement. The hazard of ending a game session is 8.7% higher for the full-crowdsourcing group than the submission-only group. When comparing the full-crowdsourcing group with the access-only group (model 2), we find that the word-submission feature does not induce a statistically significant effect on user engagement. From model 3, we further see that the addition of access to crowdsourced words does not influence user retention. Similarly, the inclusion of a word-submission option on top of the content-access option does not significantly affect user retention, as evidenced in model 4. In sum, these sets of treatment group tests indicate that there is a greater hazard of reducing session duration when an access option is provided on top of a submission option (model 1), which is aligned with the findings from models 1 and 2 of Table 5 that depict a negative interaction effect between word submission and content access. We provide detailed analysis and theoretical explanation for this seemingly counterintuitive finding in the next section, where we try to dig deep and excavate the underlying mechanisms of the impact of crowdsourcing features.

5.4. Heterogeneous Effects

It is plausible, based on the theoretical rationales provided in the hypothesis development section (Section 3), that the crowdsourcing features may have differential impact across different users. The large dispersion of the outcome variables (Table 2) also suggests that the possibility of heterogeneous effects across users. In this section, we explore whether heterogeneous impact of crowdsourcing exists on different users.

5.4.1. Heavy vs. Casual Users. In our mobile game app context, heavy users are likely to exhibit different behaviors compared with casual users. We segment the users into heavy and casual users by their total play time, which is calculated by adding up the session durations of each user. However, segmentation based on an ordering of total play time across users might be subjected to selection issues. Since the total play time, a derivative of session duration, is affected by the crowdsourcing treatment, users with high play time would naturally include more subjects from the treated group than from the control group, creating a non-randomized user segment. To circumvent this selection issue, we separately rank the total play time within each experiment group instead of pooling all the users together for the ranking. By doing so, the user segments are distributed equally among four experiment groups. We then split each of

Figure 4. (Color online) Treatment Effects on Cumulative Hazards of Abandoning the Game App



the experimental groups into two segments based on the users' total gameplay rank, where the top 25% of users in each experiment group are labeled as *heavy users* and the lower 75% percent users are labeled as *casual users*.

The heterogeneity analysis results are reported in Table 8. The submission option significantly improves the engagement levels of both heavy users and the casual users. However, the negative interaction effect of the crowdsourcing suboptions only affects the engagement of the heavy users (see models 1 and 2). Both crowdsourcing features improve the retention level of the heavy users, but their joint effect has a negative influence on user retention of the heavy users (see models 3 and 4). These results are found to be consistent under the OLS specification as well (Table B.1.4). A consistent trend across these tests is the positive impact of the submission feature and that the negative joint effect of the two features manifest solely through the heavy users and not the casual users. These results partially support Hypothesis 4a (with the exception of the submission option influencing the engagement of both user types) in that the effects of crowdsourcing elements would affect user engagement and retention of heavy users more. Despite the fact that the hypothesis is supported, it is intriguing to see that the negative influence of the joint effects on user engagement and retention uncovered earlier in Section 5 manifests mainly on heavy users. A possible explanation to unify these observations is that custom words and community words have opposing effects, especially on heavy users. Given that the amount of game time is limited within each gaming session, the set of playable crowdsourced words is split between custom and community words. If a community word appeared in a session, it would decrease the chance of a custom word appearing in the same session.[25]

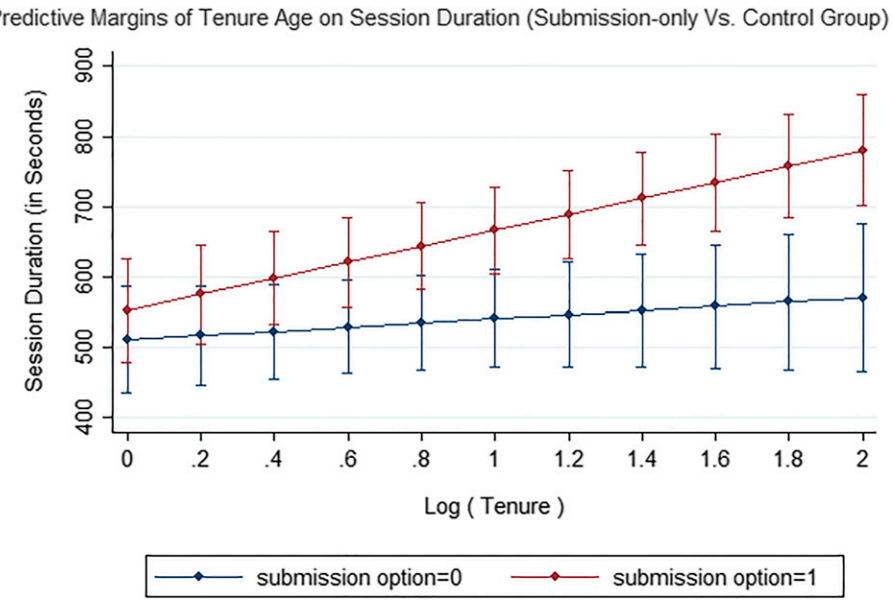
While exposure to custom words produces a greater perceived sense of autonomy to the user, exposure to community words has the opposite effect as it depletes the chance of playing the users' own custom words. Having invested more in the game, heavy users are also the ones to be content creators who submit words to the game. As such, a greater exposure to community words can diminish the level of autonomy perceived by heavy users, thereby causing a negative interaction effect on engagement and retention levels.

We tested this explanation empirically by categorizing the exposure to community words into three levels: (1) zero exposure (reference group), (2.) exposed to median number of community words (coded as a low level),[26] and (3) exposed to greater than median number of community words (coded as a high level); we investigated the interaction of this exposure and the submission option. Regressions were carried out separately for heavy and casual users, where only the interaction terms for the heavy users are significant (Table B.1.5). This implies that heavy users with the submission option, relative to the casual users, are more negatively affected by their exposure to community words. This set of results supports the notion that heavy and casual users value in-game autonomy differently. Our results can be associated with the aforementioned experiment of Ascarza et al. (2016), where the authors reported

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Figure 5. (Color online) Simple Effects of the Interaction Between Submission and Tenure



that the negative impact of the menu-based pricing strategy only applied to a consumer segment with certain use pattern, while the other segment react positively to the consumer-empowerment pricing strategy. This gives the practitioners an insight into the balance between the heterogeneous impacts that a consumer-empowerment strategy could have on different consumer groups.

5.4.2. Impact over Tenure Age. We further explore another dimension of user heterogeneity, based on users' tenure age. Because user tenure is correlated to user retention, an analysis of the effects of tenure on lifetime would be moot. Thus, we focus our analysis of user tenure only for the engagement outcome. In exploring the moderating effects of user tenure, we ran two separate models, one for each of the main crowdsourcing features. More specifically, we examine the interaction effect of submission option and user's tenure in a sample containing observations from the control and submission-only groups and investigate the interaction effect of the content-access option and users' tenure for observations from the control and access-only groups. Results of this analysis are reported in Table 9.

In model 1, we see that the interaction term of submission option and tenure is positive and significant. This indicates that the positive impact of submission option on session duration is heightened as a user's tenure in the game increases, suggesting that the content co-creation feature plays a greater role on user engagement as users mature. This effect can be seen more clearly in the plot of the simple effects (Figure 5). On the other hand, we see that the access option does not bear a significant interaction effect with tenure (Table 9, model 2). This is perhaps not surprising given that access to crowdsourced content does not have a significant impact on user engagement in the first place. These results support Hypothesis 4b, which states that the effects of crowdsourcing features are heightened, especially for users with longer tenure.

6. Discussion of Findings and Mechanisms

Thus far, we have shown the main and interaction effects of the submission and content-access features of crowdsourcing. Next, we perform in-depth analyses to understand the mechanisms behind the observed results and explore potential heterogeneous effects. In this section, we provide two streams of analyses. We first try to explain the mechanism underlying the negative interaction effect of co-creation and content access. Following that, we identify the path through which the impact of crowdsource features take place. Of particular relevance here, we assess if its impact acted specifically through the use of the feature or simply through the exposure to the option without actual word submission.

6.1. Investigating the Negative Interaction Effect

We were intrigued by the negative interaction effect on user engagement from Table 5, so we went on to investigate this nonintuitive finding further to understand

Table 8. Cox Hazard Model Estimation Results on Different User Segments

	Ending a session		Abandoning the app	
	(1) Heavy users	(2) Casual users	(3) Heavy users	(4) Casual users
Treatment received				
submission_option	0.864**	0.841**	0.576***	0.947
	(0.063)	(0.047)	(0.080)	(0.068)
access_option	1.021	0.934	0.683**	0.928
	(0.075)	(0.051)	(0.092)	(0.067)
submission_option × *access_option*	1.224**	1.056	1.611**	0.991
	(0.122)	(0.081)	(0.315)	(0.100)
Controls added	Yes	Yes	Yes	Yes
Log likelihood	−10,908.046	−19,694.296	−2,464.3219	−10,791.955
χ^2 (df)	72.24***	23.47**	30.39***	13.17*
	(12)	(12)	(8)	(8)
Observations	30,432	6,480	1,695	2,745
Subjects	1,694	2,745	747	2,309

Note. The same set of controls from the main analyses are used in all specifications.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

what might be producing this outcome. After digging deep into the marketing literature, we found that a subtle repercussion of the co-creation strategy may be responsible for the unexpected lowering of user engagement. We postulate that the negative interaction effect may be attributed to two underlying mechanisms:

First, by giving users the option to co-create content, the full-crowdsourcing version not only empowers the users to take more control of their product, but also, ostensibly, makes users more aware of their ability to control the product. It may trigger a more deliberate evaluation process. Ascarza et al. (2016) discuss this in the case of a telecommunication company proactively attempting to retain consumers by providing them a flexible menu-based pricing option. To their surprise, they found that churn rate increased among consumers who were exposed to but did not adopt the option. Reflecting on this phenomenon, the authors postulate that the flexible pricing option, which shifts more control of service use to the consumers, unexpectedly activates the consumers' originally suppressed evaluation of their current plan. This, in turn, increases their price sensitivity and churn risk. By similar logic, in our case, by giving users the ability to submit content, it is conceivable that we trigger the users' evaluation of their ability to control their product experience. Yet, and at the same time, we make them more sensitive to the loss of control. Second, when users are fed with not-quite-relevant community content through the additional content-access option compared with users whose sense of control is suppressed (do not have the co-creation option), the empowered users are more likely to feel a deprivation of control, hence lowering their engagement level. The net effect we observe is the composite of these interlaced forces.

Such activation of mental status among empowered consumers is also discussed in Wathieu et al. (2002). The authors posit that the empowerment option that triggers the process of thinking and elaborating in consumption makes consumers weigh information differently than they naturally do. Prior research supports the idea that the co-creation option raises the consumers' expectations of the product quality (Wathieu et al. 2002, Fuchs and Schreier 2011). If the expectation is not reached, the empowerment process could in turn lead to dissatisfaction on the consumer side. Considerable literature addresses this unfulfilled expectation gap issue in the context of empowerment (Broniarczyk and Griffin 2014, Pires et al. 2006). In our experiment, allowing users in the full-crowdsourcing group to access crowdsourced words

Table 9. Interaction Effect of Treatment and Tenure Age on Session Duration

	Session duration (in seconds)	
	(1)	(2)
Treatment received		
submission_option	55.697	
	(53.742)	
access_option		4.163
		(49.052)
lg (Tenure)	18.021	18.686
	(29.871)	(27.317)
submission_option × *lg (Tenure)*	78.489**	
	(37.877)	
access_option × *lg (Tenure)*		32.599
		(35.803)
No. of observations (no. of sessions)	2,145	2,103

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

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Table 10. Impact of Word Submission on User Engagement and User Retention (Cox Hazard Specification)

	User engagement hazard ratio to end the session			User retention hazard ratio to abandon the app		
	(1) Word submission vs. control group	(2) Word submission vs. no submission (full and submission-only groups)	(3) No submission vs. control group	(4) Word submitters vs. control group	(5) Word submitters vs. nonsubmitters (full and submission-only groups)	(6) Nonsubmitters vs. control group
submission_option			0.956			0.872**
			(0.037)			(0.048)
word_submission	0.652***	0.664***				
	(0.075)	(0.074)				
word_submitter				0.563**	0.652**	
				(0.108)	(0.123)	
Controls included	Yes	Yes	Yes	Yes	Yes	Yes
Log likelihood	−7,211.614	−15,985.827	−22,750.715	−3,233.630	−6,544.376	−10,081.253
χ^2 (df)	48.33***	74.71***	82.81***	22.29	19.80**	30.02***
	(10)	(10)	(10)	(6)	(6)	(6)
Observations	10,304	20,115	25,787	1,173	2,337	3,176
Subjects	1,172	2,336	3,176	832	1,540	2,222

Notes. The same set of controls from the main analyses are used in all specifications, that is, app source, geographic location, number of players, weekend indicator, and user tenure are included in the User Engagement regressions. App source and geographic location are included in the User Retention Regressions. Word Submission is a dummy variable that indicates whether the player has submitted a word in the current session. Word Submitter is a dummy variable that indicates whether the player has submitted a word in his lifetime.
*$p \le 0.10$; **$p \le 0.05$; ***$p \le 0.01$.

from the entire user community is likely to reduce their chance of seeing their own submitted words. This is because the app randomly draws each word from a word bank made up of the submitted words by both the focal user and the entire user community. When users see community words crowd out their own submitted words, the deviation between their expectation for the customized content and the actual delivery results in dissatisfaction.

We empirically validate this crowding-out effect via a logistic regression. The result in model 1 of Table B.1.2.1 shows that the likelihood of seeing one's custom words is higher among users in the submission-only condition relative to those in the full-crowdsourcing condition (the odds ratio of seeing one's own submitted words in the submission-only group is approximately 1.85 times that of the full-crowdsourcing group). Thus, it is very plausible that the chance of seeing one's submitted words decreases as the size of the community words increase. We further test this idea by exploring the reaction of seeing custom words and seeing community words among the word submitters. In our test, we replace the treatment regressors in model 1 of Table B.1.1 with number of custom words and number of community words and include only word submitters in the sample. The result in model 2 of Table B.1.2.1 reveals that seeing their own submitted words greatly enhances the engagement level, significantly increasing the session duration by approximately 821 seconds, or 14 minutes on average. On the contrary, seeing community words submitted by other people negatively affects

Table 11. Estimation of Word Submission Effect Using Instrumental Variable Approach

	(1) First-stage dependent variable: *Word Submission*	(2) Second-stage dependent variable: *Log (Session Duration)*	(3) First-stage dependent variable: *Word Submission*	(4) Second-stage dependent variable: *Log (Lifetime)*
word_submission		3.529**		4.007***
		(1.552)		(1.383)
submission_option	0.038***		0.048***	
	(0.006)		(0.008)	
Controls included	Yes	Yes	Yes	Yes
Observations (no. of sessions)	2,144	2,144	1,499	1,499

Note. The same set of controls from the main analyses are used in all specifications.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

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the word submitters' engagement level, although the result is not statistically significantly because of the small sample size. To rule out the alternative possibility that users have a greater chance of seeing their own words in a longer game session, we further ran a regression model with "Time Elapsed" as a regressor and "Saw Custom Words" as a dependent variable. The result in model 3 of Table B.1.2.1 suggests that time elapsed in a game session does not lead to a higher probability of users seeing their own custom words within a game session. In summary, the intertreatment group comparisons helped provide empirical evidence supporting the proposed crowd-out effect. However, please keep in mind that word submission is a rare event due to the nature of voluntary user content generation, and with the relatively small sample size, we take this evidence with caution.

6.2. Mechanism Underlying Feature Use

Given the previous findings, we then try to trace the impact of crowdsourcing back to feature use. More specifically, we ask the question whether the impact of crowdsourcing features is channeled through actual feature use or whether mere exposure to the crowdsourcing feature is sufficient to spark change in user behavior. In other words, we tend to distinguish between the effect of *treatment compliance* and the effect of *intent to treat*.

6.2.1. Mechanisms Underlying Word Submission. We first examine and contrast the outcomes where users participated in actual word submission with outcomes where users did not have the option and outcomes where users did not do so even though they had the option. Specifically, in Table 10, we attempt three different analyses to understand the impact of word-submission behavior on user engagement and user retention, respectively. First (in model 1), we consider the subsample of game sessions in the submission-only and full-crowdsourcing groups, which have at least one word-submission event, along with sessions from the control group. This analysis allows us to understand the effect of actual word-submission behavior on user engagement against the control group. Second (in model 2), we compare the sessions during which at least one word is submitted with sessions where there are no word-submission events among all users who are given the submission option (i.e., the full-crowdsourcing and submission-only groups). This test allows us to assess the difference between sessions with word-submission behavior and those without (complier versus noncompiler). In this analysis, we use the dummy variable *Word Submission* to indicate whether a user has submitted at least one word during a game session.[27]

In model 1, we find that the word submission is positive and significant, signifying that word-submission behavior, on average, has a positive impact on user engagement levels (reducing the hazard rate of ending the game session by 35%). In model 2, we also see that the word-submission variable bears a positive and significant coefficient. This means that sessions that involve word-submission behavior relative to those where the submission option is offered but without having actual word-submission event saw a greater increase in user engagement levels (with a 34% lower hazard rate of ending the game session). Third, in model 3, we compare the sessions that had exposure to the submission option but that did not have word-submission behavior with the control group sessions. This estimation shows that mere exposure to a submission

Table 12. Mediation Analysis at the Session Level

	Step 1	Step 2	Step 3	Step 4
	Dependent variable: *App Abandonment at Time t Cox Hazard* (hazard ratio)	Dependent variable: *Specialized Content Seeking* (OLS Coefficient)	Dependent variable: *App Abandonment at Time t Cox Hazard* (hazard ratio)	Dependent variable: *App Abandonment at Time t Cox Hazard* (hazard ratio)
Treatment received				
submission_option	0.859**	0.029		0.880**
	(0.055)	(0.021)		(0.056)
access_option	0.870**	0.041**		0.890*
	(0.055)	(0.020)		(0.057)
submission_option × *access_option*	1.107	−0.036		1.065
	(0.099)	(0.029)		(0.096)
specialized_content_seeking			0.549***	0.552***
			(0.031)	(0.031)
Controls included	Yes	Yes	Yes	Yes
Observations	4,440	4,440	4,440	4,440

Notes. The same set of controls from the main analyses are used in all specifications. The dependent variable of step 2, Specialized Content Seeking, is measured by the percentage of rounds that users choose to play the specialized topics out of the total number of rounds in a game session.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

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option does not significantly increase the session duration relative to the control. These tests are repeated under the OLS specification, in which we found similar conclusions (see models 1–3 in Table B.1.3).

We repeat the same set of analyses for retention outcomes (models 4, 5, and 6 of Table 10) under the same strategy of contrasting different subsets of observations. In model 4 of Table 10, we see that users who submit words have a lower hazard rate (by 44%) to abandon the game app compared with control-group users. In model 5, relative to users who are exposed to the word-submission feature but did not submit words, users who actually submitted words have a lower hazard rate (by 35%) of abandoning the game app in that particular game session. Despite a difference between the word submitters and non-submitters, it is interesting to see that even the mere exposure to the submission option can significantly extend user life, as shown in model 6 (with a 13% lower hazard rate to abandon the game app than the control group). These results are found to be corroborated under the OLS specifications (see models 4–6 of Table B.1.3).

Finally, to understand if the results from the earlier set of tests are influenced by the endogenous nature of word submission, we run an additional test where we use an instrumental variable approach. Our choice of instrument is based on the treatment group that subjects are assigned to. Given that treatment assignment is performed in a random fashion, the experimental group that subjects are assigned is a natural instrumental variable that satisfies the exclusion restriction. At the same time, by providing a means for subjects to submit words in the game, the treatment group with word-submission feature would naturally be correlated with the number of words submitted. Specifically, we run the following two-stage regression.

Second stage:

$$outcome_{ij} = \beta_0 + \beta_1 . wordsubmission_{ij} + Ctrl_{ij} + \varepsilon_{ij}$$

First stage:

$$wordsubmission_{ij} = \eta_0 + \eta_1 . submission_option_i + Ctrl_{ij} + \mu_{ij}$$

The outcome variables in these models are the logarithm of session duration and user lifetime in the retention regression. From the regression results in Table 11, we find that the treatment condition (i.e., submission option) satisfies the correlation requirement in that it is significantly correlated to the endogenous variable of *Word Submission* (models 1 and 3). The second-stage results further show that the *Word Submission* variable has a positive and significant relationship with session duration and user lifetime, indicating that the word submission does increase user engagement and retention, thereby affirming the results we see in model 1 of Table 10.

6.2.2. Mechanism Underlying Content Access. Next, we undertook an analysis to understand the mechanism that drives the significant impact that content access has on user retention outcomes.[28] Access to crowdsourced content can pique users' interest to explore and seek out more specialized content within the game. By playing more specialized content, users may discover more novel content, which could keep the interest level of users high for a longer time, thereby lengthening their user lifetime. We explore this possibility through mediation analysis (Baron and Kenny 1986), in which we examine whether users become more inclined to seek out specialized content under the content-access option.

Within our game app, users can choose to play card decks from 36 categories. One of these categories, called "Everything," contains miscellaneous words that cover a wide range of topics. The remaining 35 categories contain words that pertain to specialized topics. For example, the "Outdoor Adventures" category contains specialized words and phrases such as "mudslide," "deep sea fishing," or "water skiing" that are specific to the particular topic. We measure the level of specialized content that users seek by noting the percentage of category rounds played during a session and throughout each user's lifetime where specialized categories are chosen. Through the mediation analyses, we are also able to assess if such gameplay behavior mediates the main relationship between the treatment status and user retention.

The results of the mediation analysis are reported in Table 12, which depicts analysis at the session level. Step 1 of the mediation analysis looks at the impact of the various treatments on user retention, Step 2 examines the impact of different treatments on the specialized content-seeking behavior, Step 3 shows the effect of specialized content-seeking behavior on user retention, and Step 4 investigates the impact of the treatments on user retention, conditional on specialized content-seeking behavior. Results in Step 1 are expected, as they are the same analyses we saw in the main analyses. We further see that only the option of access to crowdsourced content has a positive and significant impact on specialized content-seeking behavior (the percentage of specialized topic rounds played in a game session increased by 4.1%), whereas other treatments do not have this effect (Step 2). As evidenced in Step 3, specialized content-seeking behavior enhances user retention; that is, it reduces app abandonment and increase user lifetime. When added to the same regression model as the treatments (Step 4), we find that

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specialized content-seeking behavior partially mediates the effect of the access option because the access option still has a significant effect on user retention although its significance has dropped. Through these two sets of results, we see the mechanism of content access acts by changing user behavior to seek out novel content, which in turn enhances interest levels and reduces the likelihood of churn.

7. Robustness Checks

In this section, we conduct a series of robustness checks to corroborate our results using alternative measures of the dependent variable, examine the sensitivity of our assumptions, and rule out treatment contamination.

First, we substitute session duration with an alternative dependent variable that reflects total play duration to assess if our findings on user engagement are sensitive toward alternative operationalization. Second, we execute a sensitivity analysis to assess whether the treatment effect on users' hazard of abandoning the app hold under different lifetime-censoring thresholds. Third, to account for the impact of possible contamination across treatment groups, we include the interaction terms of treatment with geographic location or number of players in our regression. Finally, we examine alternative survival analysis models to see whether these models provide a better fit with the data.

To ensure that our findings on user engagement are not sensitive to the operationalization choice of the outcome variable, we rely on total play duration as an alternative outcome variable for user engagement. This alternative measure is captured by the total time spent on the game app within the user's lifetime, which is calculated by the aggregation of the session durations over users' multiple login accesses to the app. The total play duration as an outcome variable has the advantage of integrating users' short-term interest, play intensity, access frequency, and endurance of game play into a single compound measurement. The analysis result is shown in Table B.2.1. The results show that both the full-crowdsourcing and submission-only features have a significant positive impact on enhancing users' sustained engagement, whereas the access-only feature does not have a significant impact on users' total time spent on the game. Compared with measuring short-term engagement level via session duration, the effects of the full-crowdsourcing feature on users' sustained engagement is now significant, possibly because of the compound nature of the outcome variable.

We also checked whether using 34 days is an appropriate cutoff point to denote user churn behavior. We tested our regression model with threshold values of 30, 38, and 46 days, which range from the 95th to 98th percentile among all users. The regression result remains qualitatively similar to our main regression results (Table B.2.2). Although users are randomly assigned to different treatment groups, it is nevertheless difficult to prevent subjects from different experimental groups from speaking to one another, which can lead to a spillover of treatment effects. In general, geographic locations with higher user density are more likely to suffer from contamination issues, as the proximity of individuals can lead to a greater likelihood of communications taking place between users across different experimental groups. Under this possibility, locations with higher user density would exhibit systematic differences in treatment effects from locations with low user density. To account for the possibility of contamination, we interact the treatment group with geographic location and include these interaction terms in our regression model. If users were contaminated in the described fashion, we will find a significant interaction effect between treatment and geographical location. A similar check can be performed using the interaction between treatment group and number of players, as coplayers are likely to download and use the game app in the future. Given that the higher number of players increases the chances of contamination, possible contamination because of comparisons among gaming peers may produce different use outcomes (i.e., different user engagement and retention levels), especially in game sessions that are played with a large number of players. If true, the interaction term between the treatment group and the number of players would produce significant estimates. According to our regression results in Table B.2.3, neither the interaction between treatments and geographic location nor that between treatments and number of players has a significant impact on users' hazard of ending a session. Similarly, Table B.2.4 shows the interaction terms do not have a significant impact on users' hazard of abandoning the app. Thus, it is unlikely that the experiment suffers from contamination caused by high user density or large number of coplayers in a game.

Next, we replace our Cox models with different classes of survival models to see whether alternative modeling approaches might provide better fit to our data. In this exercise, we attempt the use of the proportional hazard parametric models (PH models) assuming exponential, Weibull, and Gompertz distribution for the failure times. Results from these PH models are qualitatively similar to our results derived from the Cox model. We then make estimations using the accelerated failure-time models (AFT models), which assume generalized gamma distribution for failure events. The estimated κ and σ parameter indicate that none of the nested models with exponential distribution, Weibull distribution, or the lognormal

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distribution provide a good fit with our data. Finally, we repeat our main analyses using number game rounds and sessions as alternative dependent variables for use outcomes (i.e., user engagement and retention), to which we find that the results are largely similar to our earlier findings (Table B.2.5).

8. Discussion and Conclusion

In this work, we attempt to examine how the different processes of crowdsourcing, product co-creation, and content access might be implemented as in-app features for boosting user engagement and retention. We designed a field experiment under the context of a mobile gaming app to test our hypotheses. In our field experiment, we operationalized product co-creation as a feature that allows users to create and submit in-game content, and content access as a feature that allows users to access and play with the in-game content created by other users.

The experiment revealed several interesting insights on how each of these crowdsourcing features affects user engagement and retention. We find that user engagement can be enhanced by the content-submission feature and that user retention is enhanced by both content-submission and content-access features. Taking a closer look by analyzing feature use, we find that user engagement is only enhanced when users are offered the co-creation option and actively participate in word submission, whereas being exposed to the content-submission option and crowdsourced content both promote retention. Our findings indicate that unless the user actively submits words and consciously experiences the intervention, the co-creation strategy does not translate into an instant improvement in user engagement. This is consistent with consumer-empowerment literature, where empowerment strategy is considered to yield positive long-term effect in terms of consumer satisfaction through consumers' "retrospective evaluation process" (Wathieu et al. 2002). Meanwhile, we ponder the question of why the impact of the content-access feature is not immediately reflected in the engagement improvement. We attribute this to the particular way marketing intervention functions to affect user engagement and retention. According to the past literature (Ascarza and Hardie 2013, Ascarza et al. 2018), consumer engagement and retention are two parallel process that occur on different time scale. Over a consumer lifetime, the consumer's latent engagement state transitions from active to inactive, following a chained process in which the content quality intervention alters the consumer's latent engagement state and affects future use. We measure user engagement using session duration, which reflects the user's current latent engagement state, which is determined by her past use experience. The intervention in the current session will alter the user's latent engagement state and affect her future use, reducing her propensity to churn. In other words, there is an inertia in the effect of the content-quality intervention if the user is only passively exposed to the intervention but is not provoked to participate and more consciously sense the content quality.

Interestingly, and contrary to the conventional wisdom of *more-is-better*, we do not find complementary effects of these two crowdsourcing components on user engagement and retention. Instead, the joint impact of the two suboptions turns out to be negative. We give the possible explanation for this in Section 6. Here we further associate our findings with consumer empowerment to deepen the discussion. According to many consumer-empowerment studies (Hoffman et al. 2000, Wathieu et al. 2002, Fuchs and Schreier 2011), the successful implementation of co-creation strategy relies on an appropriate provision of progress cues to the consumers, which serve as a checkpoint that help orient consumers and provide a sense of progress. In our case, the event of seeing their own submitted word is one such important stimuli and progress cue that helps give feedback and guide consumers to participate further. Missing such cues and even replacing them with irrelevant content from other community members could cause "miswanting," a term discussed in Gilbert and Wilson (2000) that describes the impact of misprediction of future affective states on consumer dissatisfaction. Thus, the community content would be viewed as distracting and even disturbing when users' mindset is in a control-aware mode. Moreover, a second reason that our full-crowdsourcing mode fails to deliver its community-building role might be the anonymity of the game platform. The UGC literature emphasizes the importance of using content as the social object to establish the connection between the user and the community (Evans and Cothrel 2014), but for the UGC to truly develop a strong sense of shared purpose among the community, improve user retention, and sustain a long-term attachment, commonality among users tied together by the content must be discovered. In order to cleanly conduct our experiment, the users remain anonymous, which indicates that the background similarity and social ties are missing from such content sharing, which may explain why we do not observe the community-building role of the full-crowdsourcing features. Practitioners can learn a lesson from our experiment and have users' relevant background information in place to take full advantage of the crowdsourcing features.

Results also show that the actual act of word submission is a key underlying mechanism, through which the word-submission option increase user engagement and retention. Interestingly, mere exposure

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to the submission option also allows user retention to be enhanced. We further find that the content-access feature works by shifting users' game-play behavior in a way that they are more interested in seeking out specialized content. We further find that heavy users tend to be affected by the crowdsourcing features differently from the casual users in that the effects of crowdsourcing work more strongly on heavy users. Finally, we see heightened impacts of crowdsourcing in users with longer tenure age. We next discuss some of the theoretical contributions and practical implications of our work.

8.1. Theoretical Contributions

Although there has been a good amount of research on the topic of crowdsourcing, there is still a limited understanding of how the various processes embedded within crowdsourcing might be harnessed as product features to enhance user engagement and retention. Our work contributes to the literature of crowdsourcing in a few ways by assessing and quantifying the causal impacts of its subprocesses on user engagement and retention.

First, our work finds that user engagement is only enhanced by the word-submission treatment and not by content access. While crowdsourced content does produce novel content to the product offering, we theorize that the impact only shines through in the long term and thus is more evidently reflected by improved user retention. When provided jointly with a content-creation feature, the seemingly benign feature of offering access to crowdsourced content can become detrimental to user engagement, as it interferes with the user's ability to play their own words. Thus, instead of fulfilling a community-building role, the presence of crowdsourced content can in fact lower the perceived in-game autonomy. This finding constitutes as an important theoretical insight to the crowdsourcing literature, as no previous literature has examined how co-creation and content sharing should be coordinated in crowdsourcing deployment from the perspective of consumer empowerment. With the advancement of consumer-enabling IT technology, consumer empowerment through online co-creation has been thriving based on the assumption that greater consumer control leads to improved choice and greater customer satisfaction. However, challenges such as complexity of choice set and information overload exist and impair consumer experience. Existing studies mostly consider the scenario of a one-to-one co-creation relationship between the consumer and the company, neglecting the fact that consumer involvement is not only with the product but also with the community. The empirical use outcomes of directly embedding community-generated content in the co-created product have never been examined. This may be because most of the co-created products examined are still physical, such as with Muji, Threadless .com, Dell Idea Storm, or LEGO Ideas. For these physical products, the crowdsourced idea implementation is complex and must go through two steps: idea submission and voting (Fuchs and Schreier 2011). However, with arising virtual co-creation scenarios that can be easily implemented, companies should think more about how crowdsourced content should be incorporated into the products without being intrusive to users' perceived empowerment. Our study also echoed the UGC literature in emphasizing that shared content should carry social ties to truly fulfill its community-building role.

Second, the mechanism of how co-creation works is rather nuanced. Although users need to participate in the actual process of content creation to enhance engagement, the mere exposure to this feature is sufficient for boosting user retention. Such a difference can be intuitively understood when we consider the time horizon in which user engagement and retention are concerned with. User engagement is a short-term measure for users' attachment to a product during a use episode, while user retention is a long-term measure of users' loyalty to a product, which spans multiple use episodes. With a short temporal horizon to influence user engagement, interaction events would need to have a direct and substantial impression on the user to create an impressionable impact on her engagement level. As such, the actual involvement in the co-creation process is necessary to instill a positive spike in engagement. On the other hand, there are multiple use opportunities in which the app can influence users' perceptions to extend their continued use. To this end, a feature capable of producing a perception of enhanced user autonomy and control is sufficient to induce heightened user retention. Knowledge of the subtle difference in impact across use measurement constitutes a critical addition to the literature of crowdsourcing.

Third, our analyses indicate that the causal pathway in which content access improves user retention is through a shift in users' content-seeking patterns. Access to crowdsourced content sensitizes users to the availability of novel content within the product, generating an intention to seek out such content. As users embark on discovering more unique content within the product, the novelty of this content keeps user interest high, thereby lengthening users' lifetime. Although our study reveals a possible causal path that the content-access option act through, we note that the specialized content-seeking pattern is only but a partial mediator of the relationship between content access and user retention. It is worthwhile for future work to investigate other mechanisms underlying the content-access option, to supplement

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the current knowledge that this focal work has uncovered.

Finally, our analysis of effect heterogeneity across users has further unveiled the subtleties underlying the crowdsourcing components. Crowdsourcing components tend to have a larger effect on users with heavy use and longer tenure, that is, users who have experienced and consumed a larger volume of the products' existing content. Crowdsourcing features serve the role of gratifying these users by offering them alternative novel content that is beyond that provided by the app company. Knowledge of the users for whom crowdsourcing features have a greater effect highlights the boundaries by which crowdsourcing operates, adding to the crowdsourcing literature.

8.2. Practical Implications

In this paper, we assess the efficacy of crowdsourcing components in sustaining mobile app use from a product-design aspect, which is distinguished from the pricing strategy given in past literature. With all our findings discussed above, we consider our practical contribution in a few ways. First, we are the first to measure the causal impact of crowdsourcing features on user engagement and retention for the mobile app context. Second, the underlying mechanisms through which crowdsourcing features drive user engagement and prolong user retention serve as guidance for the design of the crowdsourcing feature. By virtue of our particular research setting, we gain a perspective of customization (consumer-empowerment co-creation) in the framework of crowdsourcing. Per the negative interaction effect of co-creation and content access, we suggest that practitioners should pay attention to close the loop when providing the co-creation option. Only when the delivered product and experience live up to users' expectations of autonomy will their subjective perception of empowerment be enhanced, which further leverages user engagement and retention.

Our findings also indicate that online co-creation improves long-term use outcomes such as retention, but it does not automatically translate into use intensity improvement. Digital platform developers may care about different marketing performance indices during different stages; those with new platforms that rely on daily active users (DAU) for fundraising may care more about engagement, while those with established platforms start to think about retaining users so that they can profit from subscribed long-term users. Thus, companies that focus on engagement should not only look at adding value to customers through this intervention, but should also gauge how an additional feature or content intervention would provoke the users' intent to participate and instantly initiate their evaluation process at the scene.

Finally, because the effectiveness of using crowdsourcing features to enhance user engagement depends largely on users' actual content submission, the success of crowdsourcing strategy largely depends on whether the company can motivate its users to participate in the crowdsourcing process. Designers would need to consider using incentives uncovered in the extant literature. For instance, Malone et al. (2010) point out three motivators—love, glory, and money—that characterize the crowdsourcing scene. Love and glory stand for scenarios in which users voluntarily contribute to the crowdsourcing platform as they find the task enjoyable and rewarding in themselves. To this end, gamification techniques, leadership boards, and badges are specific ways where these motivators can be operationalized in practice (Bista et al. 2012, Feyisetan et al. 2015). The use of monetary incentives can also spur user contribution (Burtch et al. 2018) but may not be efficient in stimulating quality contributions. App designers would need to assess these various incentives and select the one that is best suited for their context.

8.3. Limitations and Future Work

Because our research is an initial attempt to assess the causal impact of crowdsourcing on user engagement and retention and customer lifetime value, we look at the fundamental aspects of crowdsourcing in our experiment design. Future researchers may wish to consider secondary aspects such as the impact of the feedback or rating function for the submitted content, as these are important elements of crowdsourcing that induce a sense of recognition and motivation for users who submit content. Moreover, because of the casual nature of our study context (i.e., mobile game app), the crowdsourced content is largely for serving personal, hedonic purposes. Further studies could consider different contexts where crowdsourcing is used for fulfilling other purposes (e.g., in a news sharing app or banking app). In addition, because of company restrictions, we were only able to run the experiment for a limited period of time. It is unclear whether the impact of crowdsourcing features would last over longer periods. Furthermore, although we have uncovered word submission as a viable mediator of the main relationship, the methodology of our work is unable to reveal the psychological reasons to such user behaviors. It might be worthwhile for future work to explore these mental pathways via surveys. Finally, to abstract away from of the effects of social ties we chose to hide the identity of users who submitted game content in our experiment design. In future work, it would be interesting to examine the impact of crowdsourcing under a community with

known user identities. Notwithstanding these limitations, we believe our findings provide a set of baseline results and lay a good foundation for the future studies on crowdsourcing features in products.

Endnotes

[1] Figures are from https://www.statista.com/topics/1002/mobile-app-usage/.

[2] The cumulative number of app downloads reached 50 billion for Google Play in 2013 and 100 billion for Apple App Store in 2015 (http://www.statista.com/statistics/281106/number-of-android-app-downloads-from-google-play/; http://www.statista.com/statistics/263794/number-of-downloads-from-the-apple-app-store/).

[3] See http://whatsnext.nuance.com/customer-experience/95-apps-quickly-abandoned-avoid-becoming-statistic/.

[4] As of September 2015, Wikipedia attracts more than 370 million unique visitors monthly, whereas Quora boasted 100 million monthly unique visitors in March 2016. Threadless was estimated to have a $30 million sales and a 30% profit margin in 2012.

[5] In this paper, we use the terms "user," "consumer," "customer," and "player" interchangeably. All of them refer to the person who uses or consumes the mobile app. These terms are consistent in meaning and used alternatively per convention of the different body of literature we mentioned (e.g., literature in management information systems, marketing, or human computer interaction).

[6] The authors suggest that the crowdsourced content may function as a social channel when accessed by the user community and thus furthers engagement and retention.

[7] Although Ascarza et al. (2016) do not directly address the issue of consumer empowerment or co-creation, the flexible menu-based pricing option discussed in their paper is actually a typical case of consumer empowerment by shifting more choices to the consumers. They also find that this additional option that allows consumers to customize their use plan unexpectedly increases consumer price sensitivity. See Section 6.1 for more details.

[8] An instance of this is Starbucks using the crowdsourced ideas of splash sticks, new coffee flavors, and birthday treats in its operations.

[9] We are aware that such a conceptualization might not apply in certain crowdsourcing contexts in which the crowdsourcer (i.e., initiators in crowdsourcing contests) solicits ideas for internal use. Hence, we intend to apply our conceptualization strictly for crowdsourcing models that are used on products/services that would be made available publicly.

[10] User tenure is directly related to user lifetime. For this reason, it is moot to examine the moderating effects of user tenure on user retention, to which we do not hypothesize.

[11] Derived from https://www.statista.com/statistics/234649/percentage-of-us-population-that-play-mobile-games/.

[12] For the sake of simplicity, we use "words" throughout this paper to refer to both words and phrases.

[13] See https://www.vertoanalytics.com/average-mobile-game-day/ and http://www.businessofapps.com/gameanalytics-report-reveals-the-gap-between-top-performing-mobile-games-and-the-rest-reaches-an-all-time-high/.

[14] See https://techcrunch.com/2017/06/21/mobile-gaming-sessions-down-10-percent-year-over-year-but-revenue-climbs/ and https://techcrunch.com/2017/01/12/the-mobile-app-gold-rush-may-be-over/?_ga=2.47056100.2104755912.1539705559-423448281.1539705559.

[15] See https://www.forbes.com/sites/johnkoetsier/2017/10/20/the-very-best-android-game-has-just-4-5-consumer-retention-after-30-days/#6fe4e0cc12fb).

[16] By using this set of sample sizes, we are able to comfortably detect a small effect at the conventional alpha and power levels under a factorial design analysis.

[17] There are alternative measures of user engagement that may be deployed in our study context. We included tests involving such variables as sub-robustness tests (results shown in Table B.2.5).

[18] This measurement of user engagement in the mobile app context is similar to that used by Zhang et al. (2019), which relies on the amount of time spent reading eBooks in a mobile reading app to measure users' engagement with the reading app. See Appendix A.3 for details on the game rounds and sessions.

[19] A test for equality of proportions also reveals that the proportion of users is similar, which further shows that our randomization procedure is successful. We thank an anonymous reviewer for pointing this out.

[20] By design, we do not conduct a manipulation check for the access feature. To assess whether users perceive the in-game crowdsourced words would involve the intrusive solicitation of user responses. Such an act would make users realize the existence of the experiment, which can then introduce unwarranted demand effects. Considering the tradeoff of invalidating the field experiment, we decided to mitigate concerns of non-perception by enhancing the saliency of access feature via the app design.

[21] The 50% is calculated by subtracting the mean consumer lifetime of the control group users from that of the full-crowdsourcing group users and dividing by the mean consumer lifetime of the control group users, which results in $(7.769 - 5.165)/5.165 = 50\%$. Similar formulations also apply to the calculation of the 43% and 31% increments.

[22] Given that our experiment is limited to a three-month period, the observed user lifetime of subjects who joined the experiment near the end of the third month is highly likely to be censored.

[23] Specifically, we sampled 534 subjects from each condition. This number of subjects is twice the amount recommended under Cohen's calculation. We have chosen to use a larger sample size for this analysis for similar reasons that were cited earlier on alleviating concerns unique to our study context.

[24] Despite the same regressors having the statistical significance, we note their significance level has dropped slightly as a result of a reduced sample size.

[25] We verified this empirically using our data set. See description in Section 6.1.

[26] The median number of community words users are exposed to during a game session is 1.

[27] Figure A.7.1 illustrates the point of estimation, as well as our coding of the word-submission variable along the game-playing timeline.

[28] We do not look at the mechanism of content access on user engagement because the main analyses did not reveal a significant relationship between the two.

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